

THE INFLUENCE OF LIFESTYLE PRACTICES AND THE VULNERABILITY TO
RESPIRATORY DISEASES OF PERSONS DEPRIVED OF LIBERTY AT
DAVAO ORIENTAL PROVINCIAL JAIL



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BACHELOR OF SCIENCE IN NURSING

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**DAVAO ORIENTAL
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APPROVAL SHEET

This thesis proposal hereto attached entitled **"THE INFLUENCE OF LIFESTYLE ON THE VULNERABILITY OF PERSONS DEPRIVED OF LIBERTY TO RESPIRATORY DISEASES AT DAVAO ORIENTAL PROVINCIAL JAIL"**, prepared and submitted by **FLORIELYN JOYCE P. APARECIO, ROSHABELLE C. ACUÑA, ELLAH MARIZ T. MAPA, AND KAYE P. VILLANIZO**, is hereby recommended for approval and acceptance.

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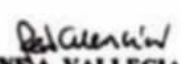

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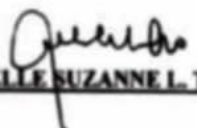
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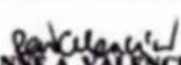

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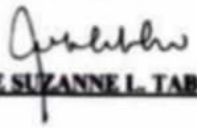
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ABSTRACT

FLORIELYN JOYCE P. APARECIO, ROSHABELLE C. ACUÑA, ELLAH MARIZ T. MAPA, KAYE P. VILLANIZO (2025). THE INFLUENCE OF LIFESTYLE PRACTICES AND THE VULNERABILITY TO RESPIRATORY DISEASES OF PERSONS DEPRIVED OF LIBERTY AT DAVAO ORIENTAL PROVINCIAL JAIL. Bachelor of Science in Nursing

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The prison system is a breeding ground for the contraction and transmission of a variety of diseases (Barvo et al., 2021). These conditions pose a heightened risk for the spread of infectious diseases within these facilities (Novisky et al., 2021). The lack of information regarding PDLs' vulnerability to respiratory diseases is a cause for concern, as it impedes the development effective public health interventions and preventive measures. This study aimed to assess the influence of lifestyle practices and the vulnerability of persons deprived of liberty at Davao Oriental Provincial Jail, using a quantitative, descriptive–correlational, and comparative design. The survey questionnaire was pilot-tested and evaluated using Cronbach's Alpha (0.833). A total of 62 respondents participated in the study, with the majority aged 26–35 and 36–45 years, each group comprising 33.9% of the sample. The respondents included 50 males (80.6%) and 12 females (19.4%). Data were collected using an adapted questionnaire assessing hygiene, diet, physical activity, and health vulnerability. Findings revealed an overall excellent level of lifestyle practices (mean = 4.51), with hygiene scoring highest (mean = 4.89), followed by diet (mean = 4.50) and physical activity (mean = 4.15). The vulnerability level was low (mean = 3.81), reflecting the excellent outcomes of the lifestyle practices. Pearson correlation analysis showed a significant relationship between lifestyle and vulnerability ($r = 0.305$, $p = 0.016$), particularly in diet ($r = 0.306$) and physical activity ($r = 0.283$), which indicates that as the quality of life improves, the vulnerability to respiratory diseases tends to decrease. These results underscore the need for targeted interventions and health education to mitigate the prevalence of respiratory illnesses at the Davao Oriental Provincial Jail. The study concludes with recommendations for nursing education, practice, and future research.

Keywords: *Hygiene, diet, physical activity, correctional facility*

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CHAPTER I

INTRODUCTION

Background of the Study

Populations behind bars have higher rates of acquiring diseases, which are made worse by restricted access to preventative care. The prison system is a breeding ground for the contraction and transmission of a variety of diseases (Barvo et al., 2021). The physical environment and conditions within correctional facilities inherently contribute to a highly pathogenic setting. Factors such as poor sanitation, inadequate ventilation, and the impossibility of practicing social distancing due to overcrowding create an environment conducive to the increased transmission of highly contagious viruses within carceral setting.

These conditions pose a heightened risk for the spread of infectious diseases within these facilities (Novisky et al., 2021). Persons Deprived of Liberty (PDLs) in countries with lower to moderate income levels face respiratory health risks due to unfavorable conditions in prisons (Eze et al., 2022). In Peru, 6,416 incarcerated individuals, predominantly men, were diagnosed with chronic lung conditions, while 3,267 PDLs, again mostly men, had infectious or contagious lung diseases such as tuberculosis (Iglesias-Osores, 2020). The Philippines, known for having one of the most crowded prison systems globally with over 215,000 PDLs, faces issues like severe overcrowding, inadequate ventilation, and high disease rates among PDLs (Cruz, 2022). The lack of movement, limited access to physical labor, and inadequate dietary intake contribute significantly to their declining health (Kosendiak et al., 2022).

It is widely acknowledged that lifestyle factors impact the onset and prognosis of various illnesses (Murano et al., 2023). Diet and nutrition, modifiable factors, play a role in infectious diseases' acquisition and severity (Vu et al., 2022). Incarcerated individuals face challenges in maintaining a healthy training routine, potentially due to the absence of adapted sports in prison. Also, the study of Pageau et al. (2021) reveals dissatisfaction with the nutritional aspects of prison life, with participants expressing concerns about the mediocre taste and poor quality of meals. Fatty ingredients are reported to be excessively present in the diet, despite awareness of their potential harm to health. Due to the large inmate population, food supplies are carefully rationed to ensure all prisoners receive a share. Many inmates report facing various difficulties, with some highlighting inadequate food portions and limited water supply as major concerns. The rising number of detainees has led to insufficient food distribution, worsening the situation (Gales et al., 2023).

Many studies have investigated the factors affecting the vulnerability of PDLs to diseases. However, there is limited research on the influence of lifestyle practices such as hygiene, diet, and physical activities and level of vulnerability to respiratory illnesses on the respondents. Furthermore, existing studies on respiratory diseases predominantly concentrate on COVID-19, overlooking other critical conditions like pneumonia, chronic obstructive pulmonary disease, and related respiratory ailments. Moreover, existing studies rarely examine the influence of factors such as sex, age, ethnicity, educational attainment and level of vulnerability to respiratory diseases. The lack of such information regarding PDLs' vulnerability to respiratory diseases is a cause for concern, as it impedes the development of effective public health interventions and preventive measures within the prison system. Understanding the influence of lifestyle factors and the level of vulnerability of persons deprived of liberty to respiratory diseases is crucial for devising effective interventions within correctional facilities.

This study addresses existing research gaps by evaluating the influence of lifestyle practices and the level of vulnerability to respiratory diseases of the respondents. Recognizing this impact is essential for crafting tailored interventions and support initiatives, promoting rehabilitation, and ensuring equitable and efficient operations within the criminal justice system.

Statement of the Problem

This study sought to determine the influence of lifestyle practices and the vulnerability to respiratory diseases of the respondents. This specifically aimed to answer the following questions:

1. What is the demographic profile of the respondents in terms of:
 - 1.1 Sex;
 - 1.2 Age;
 - 1.3 Ethnicity;
 - 1.4 Educational Attainment; and,
 - 1.5 Frequency of acquiring respiratory diseases?
2. What is the level of lifestyle practices of the respondents in terms of:
 - 2.1 Hygiene;
 - 2.2 Diet; and,
 - 2.3 Physical Activity?
3. What is the level of vulnerability to respiratory diseases of the respondents?
4. Is there a significant relationship between the respondents' level of lifestyle practices and their level of vulnerability to respiratory diseases?
5. Is there a significant difference between the respondents' level of vulnerability to respiratory diseases when grouped according to demographic profile?

6. Does lifestyle practices in terms of hygiene, diet, and physical activity predict the level of vulnerability to respiratory diseases of the respondents?

Hypotheses

Ho1: There is no significant relationship between the respondents' level of lifestyle practices and their level of vulnerability to respiratory diseases.

Ho2: There is no significant difference between the respondents' level of vulnerability to respiratory diseases when grouped according to demographic profile.

Ho3: The lifestyle practices in terms of hygiene, diet, and physical activity does not predict the level of vulnerability to respiratory diseases of the respondents.

Significance of the Study

The study on the influence of lifestyle practices and the vulnerability to respiratory diseases of the respondents carries substantial implications for multiple facets of the nursing profession.

Nursing Education. The study's findings can be integrated into curriculum enhancement. By incorporating these elements into the nursing curriculum, educators can better prepare future nurses to address the unique health needs of persons deprived of liberty.

Davao Oriental Provincial Jail Administration. Findings from this study can inform the management of correctional facilities, enabling them to implement more effective measures to reduce disease transmission and improve healthcare within the jail. The research can contribute to evidence-based policymaking and the development of interventions tailored to the specific needs of PDLs.

The Persons Deprived of Liberty. This study can shed light on how their lifestyle choices affect their vulnerability to respiratory diseases. Understanding these influences

can empower them to make healthier choices and take preventive measures, potentially reducing their risk of respiratory illnesses while incarcerated.

Nursing Practice. The study can be used to promote a more comprehensive assessment by considering lifestyle factors when caring for persons deprived of liberty. It can also encourage nurses to design interventions that address the specific respiratory health needs of PDLs, considering their living conditions and lifestyle.

Future Researchers. The study can be a basis for expanding research on the broader health implications of incarceration and lifestyle on various health conditions beyond respiratory diseases.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter is about the related studies or literature connected to this study. This part of the study discusses certain topics related to this study: the lifestyle of persons deprived of liberty, persons deprived of liberty's vulnerability to diseases, the influence of lifestyle to vulnerability, healthcare access, health education, and preventative measures. The articles used in this study would be from journals, websites, news reports, and research studies found on the internet.

Lifestyle of Persons Deprived of Liberty

Lifestyle was defined by Bell (2021) as the unique way in which an individual or a group fulfills their biological, economic, emotional, and social needs, often mirroring their attitudes, beliefs, and values, constitutes their particular lifestyle. In this study's context, the deprivation of liberty, due to its attributes, leads individuals to develop specific behaviors and traditions that may impact their well-being (Costa et al., 2023).

Moreover, there has been a disregard for physical exercise within global prison systems for years. Overcrowded prisons frequently foster unhealthy environments, and limited opportunities result in a lack of physical activity for persons deprived of liberty (Baldonado et al., 2022).

Additionally, in the Philippines, the designated budget for the food of the PDLs stayed fixed at seventy (70) pesos per day, while fifteen (15) pesos was allocated for medical expenses. It is twenty (20) pesos higher than in 2015, wherein the PDLs' budget for food is fifty (50) pesos (Bolledo, 2023). This problem is not just in the Philippines, as

per Burki (2020), prisons in DR Congo operate at an estimated 432% of their intended capacity, yet the food budget is based on the official capacity, resulting in provision for a maximum of just one meal per day. In Niger, more than half of the prison population, specifically those in pretrial detention, receive no food provisions.

Furthermore, continuous monitoring of adequate nutrition in prisons is essential due to limited external oversight and a meager budget allocated for meals, along with unclear requirements in this aspect. A research study in Poland found out that the dietary intake of minerals fell short of meeting the recommended dietary allowance (RDA) for calcium and magnesium. Of concern was the excess intake of sodium and phosphorus in the standard diets. Vitamin deficiencies were most notable in vitamins D, C, and folate. Conversely, an overly high intake was observed for vitamins A and B12. The findings of this study highlight the necessity for more precise legal regulations governing nutrition in Polish prisons. These regulations should encompass not just energy value and macronutrient provision but also the intake of minerals and vitamins (Stanikowski et al., 2020).

Subsequently, the increasing occurrence of contagious illnesses in prisons is largely caused by inadequate sanitation and poor hygiene practices. In a study conducted by Engdaw et al. (2023), PDLs' hygiene is influenced by factors like cell crowding, water access, and knowledge level. Over half had good hygiene practices, with knowledge, water intake, and cell density playing key roles. Improving water access is crucial for better hygiene, alongside educating PDLs to prevent disease spread.

In addition, according to Eze et al. (2022) inmates are at increased risk of illness due to harmful conditions in correctional facilities, including overcrowding, poor ventilation, indoor air pollution, inadequate nutrition, and limited healthcare access. While respiratory infections like tuberculosis are well documented, data on non-communicable respiratory diseases (NCDs) in prisons is lacking. However, research

suggests that poor indoor air quality contributes to NCDs such as asthma, COPD, lung cancer, and reduced lung function. Indoor tobacco smoking, poor ventilation, and limited outdoor access worsen exposure to harmful air pollutants.

Persons Deprived of Liberty's Vulnerability to Diseases

Persons deprived of liberty are particularly vulnerable group of people. Compared to the general population, they have higher healthcare demands and are typically disadvantaged. Due to the jail's population that fluctuates by about 65,000 people annually, the health problems of PDLs affect the entire community (AIHW, 2022). The Australian Institute of Health and Welfare previously stated that individuals incarcerated in prisons represent one of the most susceptible groups within society, often originating from underprivileged backgrounds. Those who have been incarcerated tend to encounter elevated occurrences of homelessness, joblessness, mental health issues, persistent physical ailments, contagious diseases, tobacco usage, excessive alcohol consumption, and illicit drug use in comparison to the wider population.

Also, in the Philippines, there were 477 prisons as of June 2022. Over 131 thousand people were incarcerated throughout the nation, including both detainees and those condemned to prison (Balita, 2023). As stated by De Leon (2023), the New Bilibid Prison, in particular, is 477% full as of June 30, 2023, with a population of 30,701 persons of deprived liberty. The facility can only accommodate 6,345 inmates. The highest ratio recorded was 1:57 in 2019, just before the COVID-19 pandemic began. It was 1:9 in 1990. The Manila City Jail, popularly known as Old Bilibid Prison, is hailed as one of the most overcrowded prisons in the world.

In addition, the Philippines faces an exceptionally difficult situation due to having the highest jail occupancy rate globally. Overcrowded prisons and limited resources make PLDs and detainees exceptionally vulnerable to COVID-19. PDLs and detainees

are described as “hidden victims” of the pandemic due to the lack of attention and resources in prisons. Overcrowding, lack of nutrition, and inadequate safety measures contribute to their vulnerability (Kahambing, 2021).

Similarly, in African prisons, there is an urgent need for improving the health and well-being of prisoners and prison staff. The author emphasizes the severe health disparities in African prisons, including overcrowding, lack of sanitation, malnutrition, and inadequate healthcare. It mentions that some prison conditions are life-threatening, as observed by international human rights monitors. Prisons are also identified as environments conducive to the spread of diseases such as HIV, TB, hepatitis, influenza, and COVID-19. The prevalence of these diseases among prisoners is much higher than in the general population (Van Hout, 2020).

Additionally, in Brazil, persons deprived of liberty face a heightened risk of contracting COVID-19 attributed to cramped and poorly ventilated cells, limited access to basic sanitation, and the absence of health facilities (Lins & Rocha et al., 2021). A recent ecological study examined the five macro-regions of Brazil, including the Federal District. Findings revealed 14 geographic areas with heightened COVID-19 risk among individuals in detention, with the Federal District identified as the cluster with the highest risk (De Almeida Crispim et al., 2021).

Moreover, in the study of Herrera et al. (2022), they emphasized that factors like overcrowding contribute significantly to the spread of the disease, fostering increased closeness and prolonged contact between active TB cases and vulnerable individuals. Additionally, delayed diagnosis and initiation of TB treatment, restricted air circulation, elevated humidity, minimal exposure to ultraviolet (UV) light, and host-related factors further influence this transmission. Furthermore, Swarthout et al. (2020) stated that maintaining good hygiene and access to safe drinking water could minimize pathogen

exposure and boost immune responses, which lowers the incidence of acute respiratory infections.

Additionally, the study of Ashraf et al. (2020) shows the efficacy of the handwashing intervention, which involved promoting homemade soapy water with free detergent refills and providing free handwashing stations close to the kitchen and latrine. The results of this study add to the body of research showing that sanitation programs reduce childhood respiratory illnesses. It is probable that the children benefited from decreased respiratory sickness due to stronger immune systems and adequate vitamin levels, as the researchers saw a significant reduction in diarrheal infections in the sanitation arm of this trial.

The Influence of Lifestyle to Vulnerability

In terms of physical activity, Mohamed et al (2020) conducted a review that revealed enhancing aerobic capacity could lead to short-term, safe enhancements in immune and respiratory system functions, particularly beneficial for combating COVID-19 infections. This improvement is attributed to three main mechanisms. Firstly, it boosts immunity by increasing immune cell levels and activity, regulating CRP levels, and reducing anxiety and depression. Secondly, it enhances respiratory functions by exhibiting antibiotic, antioxidant, and antimycotic properties, as well as restoring lung tissue elasticity and strength. Lastly, it serves as a protective barrier, lowering COVID-19 risk factors and consequently reducing the occurrence and severity of the disease.

Meanwhile, for those who are asymptomatic or just have minimal symptoms, exercise is also the most effective form of therapy. These individuals are typically vulnerable, such as the elderly and those with chronic illnesses. Therefore, continuing physical activity would decrease implications in the future from COVID-19 and lessen its comorbidities (Filgueira et al, 2021).

In conjunction, the research findings of Chastin et al. (2021) stated that engaging in frequent, moderate to vigorous physical activity improves the immune system's first line of defense, boosts vaccine efficacy, and lowers the risk of infectious disease mortality and community-acquired infections. The study of Khoramipoura et al. (2020) also states that active lifestyles activate the immune system, while obesity and related disorders suppress it, as exercise plays a crucial role in inflammatory processes, and has effects on the immune system. Intensity is the most important factor, but recent studies show no suppression effect of high- intensity interval training (HIIT) on immune function. However, moderate-intensity exercise increases immune cell count and reproduction, enhancing cytokine and cytotoxic granule production. While many studies focus on aerobics training, resistance training with moderate intensity and volume also improves immune function. In addition, concurrent training is less effective than sole aerobic or resistance training.

Moreover, numerous studies have shown that malnutrition, excess weight, and obesity have detrimental effects on the immune system, increasing susceptibility to viral infections. Additionally, various research findings indicate that dietary interventions can serve as immune boosters, potentially reducing the risk of viral infections (Moscatelli et al.,2021). In order to increase the viral vulnerability associated with malnutrition and obesity, meal choices must take into account the inflammatory effects of certain foods, as demonstrated by the study conducted by Morais et al. (2021). Probiotics, micronutrients, bioactive substances, and healthy food practices are COVID-19 preventive tactics. As a result, eating a varied and well-balanced diet can help boost the body's defenses against viral infections like COVID-19.

Additionally, vitamins, trace elements, amino acids, and fatty acids play key roles in supporting the immune system and reducing infection risk. Consuming a diverse diet of plant-based and animal-based foods is best for supporting the immune system.

However, some micronutrients, such as vitamins C, D, and E, and zinc and selenium, may be in excess of intakes that can be achieved through diet alone. The gut microbiota plays a role in educating and regulating the immune system, and probiotic bacteria can modify the microbiota and protect against infections. A healthy diet could be an important factor in determining outcomes in individuals infected with coronavirus (Calder, 2020).

Furthermore, enhancing nutrition through targeted micronutrient supplementation is a promising strategy to strengthen the immune system and reduce the severity of infections, including COVID-19—especially in low- and middle-income regions facing undernutrition and deficiencies in vitamins A, B complex, C, and D (Akhtar, 2021). Coupled with lifestyle modifications such as improved diet, regular physical activity, adequate sleep, and reduced smoking and alcohol consumption, these measures can help lower infection risk and prevent severe respiratory outcomes (Lange & Nakamura, 2020).

Healthcare Access

Human rights groups have examined the living situations within Philippine prisons and highlighted that inmates face inadequate access to food, water, sanitation, and healthcare services (Arambulo et al., 2021). In an interview to hospital medical Chief Ernesto Tamayo, approximately 5,200 prisoners pass away each year due to overcrowded prisons, unsanitary living conditions, and limited availability of basic healthcare. In addition, restricting PDLs' access to healthcare services might result in future health issues. If the health of PDLs remains unmonitored, outbreaks could emerge within detention facilities, potentially leading to a broader health crisis for the entire country (Auro, 2021).

Prison inmates are at increased risk of illness due to harmful conditions in correctional facilities, including overcrowding, poor ventilation, indoor air pollution,

inadequate nutrition, and limited healthcare access. Indoor tobacco smoking, poor ventilation, and limited outdoor access worsen exposure to harmful air pollutants (Eze et al., 2022). Additionally, all smokers reported respiratory infections, indicating that both cell features and lifestyle choices significantly contribute to respiratory health issues among inmates. These findings underscore the need for non-medical interventions to improve respiratory health management in the prison setting. In Brazil, the 2014 National Policy for Comprehensive Healthcare for Persons Deprived of Liberty (PNAISP) oversees public health strategies for the prison system. Its objective is to ensure that PDLs can avail themselves of comprehensive healthcare within the Unified Health System (SUS). However, it's crucial to contemplate how the execution of this policy is significantly influenced by the beliefs, notions, and ethical judgments of those responsible for implementing it in prison settings (Bartos, 2023).

Apart from this, incarcerated women in North America face limited access to reproductive healthcare. Research conducted separately in two prisons in Canada and the United States revealed that about 80% of incarcerated women lacked adequate access to contraception. Globally, women who undergo imprisonment have reduced access to prenatal care, leading to higher occurrences of pregnancy complications like preterm birth and low birth weight, compared to women in the general population. (Liauw et al., 2021).

Health Education

In a study by Valentim et al. (2022), the health of individuals in prisons is a growing and increasingly important subject, despite persistent neglect in many countries regarding human rights, the universal right to health, promotion of well-being, and disease prevention in specific populations. The heightened interest in prison health is driven by the worsening health issues in custodial settings, turning it into a public health

concern. The state of prison health has further worsened due to the global increase in the prison population. This rise, coupled with escalating health problems, has garnered worldwide health attention in recent decades. In such circumstances, relying solely on a continuing education program with primarily in-person instruction, aiming to enhance health system responsiveness, appears insufficient, mainly due to scalability concerns and the pressing need for educational offerings.

Furthermore, Mehay et al. (2021) found out that out of 75 young men in prison surveyed, 72% demonstrated limited health literacy. Barriers encompassed structural constraints, restricted access to formal support, and disruptions in social and environmental contexts. Limited health literacy was not predicted by demographic characteristics or smoking intentions/behaviors but was associated with prison-related factors. Additionally, physical issues (sleep, nausea, tiredness, and headaches), mental health and well-being (anxiety, depression, and affect), and somatization problems were indicative of these limitations.

Moreover, in the study of Sentell et al. (2020), health literacy is essential in addressing various impacts of the crisis across different aspects of society, from individual behaviors to international economies. The special issue focuses on two key principles: defining health literacy as essential skills and resources for people to access, understand, evaluate, communicate, and use health-related information and services; and recognizing that improving health literacy is vital for achieving health equity. Another article stated that health education can also have a positive impact on a community's economic well-being by lowering healthcare expenses and minimizing productivity losses caused by preventable illnesses. Initiatives aimed at assisting community members in addressing these costly health issues not only improve individuals' well-being but also offer a substantial return on investment for communities (Gagnon, 2022).

In addition, education regarding health holds significance for all individuals. Those within the justice system and personnel working in correctional and detention centers require heightened awareness regarding risks, prevention, and treatment methods for certain diseases and conditions. These include HIV, viral hepatitis, sexually transmitted infections (STIs), tuberculosis (TB), and latent TB infection (LTBI) (Centers for Disease Control and Prevention, 2023). Education focused on healthcare within correctional facilities has served as a policy catalyst, influencing procedural modifications adopted by the Brazilian National Health System (SUS). This emphasis is evident in ongoing health education programs, which aim to enhance healthcare services for individuals deprived of their liberty (Valentim et al., 2023).

Also, efforts directed towards the welfare of the prison population address a significant social justice concern. Various international organizations view endeavors aimed at improving health and healthcare education for individuals within prisons as essential. These efforts have been the subject of several studies and projects in recent years (Moreira, 2022). Delivering healthcare within prisons is essentially a means of upholding the human rights of individuals deprived of their liberty, as well as protecting the broader community that is directly and indirectly impacted by health issues within these settings. Therefore, prison health must take into account not only the prison facilities themselves but also the surrounding area, including inmates, their families, correctional staff, and the local community that interacts with the prison system (Valentim et al., 2022).

Preventative Measures

Preventive health services involve a range of healthcare measures designed to conduct screenings and potentially detect health issues before the onset of symptoms. Engaging in preventive healthcare can contribute to an extended and healthier lifespan

(Wisner, 2023). Primary prevention involves measures directed at a susceptible population or individual to thwart the occurrence of a disease. Its objective is to forestall the onset of a disease, focusing on a population initially in good health. This typically involves activities that either minimize exposure to risks or enhance the immunity of individuals at risk, thereby preventing the progression of a disease from susceptibility to subclinical stages.

On the other hand, secondary prevention centers on the early detection of diseases, targeting seemingly healthy individuals with subclinical manifestations. Tertiary prevention addresses both the clinical and outcome stages of a disease, implemented in symptomatic patients to mitigate disease severity and associated consequences. While secondary prevention seeks to avert the onset of illness, tertiary prevention aims to alleviate the effects of an established disease in an individual. Rehabilitation efforts are frequently employed as forms of tertiary prevention (Kisling, 2023).

Also, the World Health Organization (2023) stated that it is important to highlight that while primary prevention measures can be carried out without necessarily enhancing the capacity of other healthcare services, the same isn't true for secondary prevention. Screening and early detection have restricted effectiveness (and could even be harmful to the patient) if issues identified cannot be swiftly addressed or treated through other healthcare system services.

Furthermore, a prison should employ both "general measures" and "local measures." General measures involve actions within the prison, such as using filtering face piece 2 (FFP2s), conducting an effective vaccination campaign, population screening for SARS-CoV-2 through swabs and serology, implementing safety measures, isolation and quarantine, reducing the number of prisoners in cells, and enhancing the health and hygiene of all inmates. "Local measures" refer to actions taken in cases of PDL positivity. In instances of suspected positivity, notifying the CTT, conducting contact

tracing, symptom monitoring, quarantine, and molecular swabs for SARS-CoV-2 on close contacts are necessary, accompanied by psychological support (Esposito et al., 2022).

Moreover, for individual experiencing respiratory symptoms, offer guidance on maintaining respiratory hygiene and proper cough etiquette. Emphasize the use of tissues for disposal, covering the nose and mouth during coughing or sneezing, and practicing hand hygiene. Provide necessary materials such as tissues, disposal containers, and personal alcohol hand gel dispensers. Isolate individuals with symptoms and their close contacts until 24 hours after symptoms subside, ensuring a 3-6 feet distance if separation is not possible. Those without symptoms should refrain from close contact with sick individuals, except when essential for care and support. Evacuation center workers displaying symptoms should be excused from their duties (Centers for Disease and Control Prevention, n.d.).

Additionally, a study conducted by Talic et al. (2021) indicated that various personal protective and social measures, such as handwashing, wearing masks, and practicing physical distancing, are linked to decreased incidences of COVID-19. Maintaining physical distance and wearing face masks are unquestionably the most effective preventive measures to minimize the risk of exposure to the new coronavirus. Besides fundamental health and hygiene practices such as handwashing, the Centers for Disease Control and Prevention (CDC) (2023) advocates several preventive measures for all levels of COVID-19 hospital admissions. These include keeping up with COVID-19 vaccinations, enhancing ventilation and outdoor activities, undergoing COVID-19 testing if necessary, adhering to guidelines for exposure, staying home if there's suspicion or confirmation of COVID-19, seeking treatment for high-risk COVID-19 cases, and steering clear of contact with individuals suspected or confirmed to have COVID-19.

Also, enforcing physical distancing within prisons is challenging, with inadequate protective measures available. Failure to manage infections within prisons could impede global endeavors to contain the disease. Those in detention face heightened susceptibility to contracting COVID-19 and are particularly at risk of human rights abuses due to unjustified restrictions imposed under the guise of infection control (Pont et al., 2021). Thus, prisons must offer health consultation services and enforce smoke-free regulations to minimize the risk of acute respiratory infections (ARIs). Additionally, it's important to address waste management in prisons to mitigate the spread of respiratory diseases (Satria, 2023).

Synthesis

Research consistently shows that a person's lifestyle can greatly affect the physical health of those in prison. Many studies have found a strong link between the two. People in correctional facilities are more likely to get sick due to harsh conditions such as overcrowding, poor ventilation, indoor air pollution, malnutrition, and limited access to health care. Improving nutrition, increasing physical activity, getting enough sleep, and cutting back on smoking and alcohol could help reduce the risk of infections and prevent serious respiratory diseases.

However, it's important to note that research in this area is still ongoing, and there are gaps in our understanding. For example, fewer studies have looked at how lifestyle factors specifically affect the risk of respiratory diseases among prisoners, whose lifestyle is largely shaped by what the facility provides. While many studies focus on living conditions, diet, hygiene, and physical activity, more research is needed on this topic. In conclusion, the relationship between lifestyle and vulnerability to respiratory diseases is complex, and future research should explore possible interventions to help prisoners adopt healthier habits.

Theoretical Framework

The theoretical framework of this study is built upon the Health Lifestyle Theory. Health lifestyles refer to collective behavioral patterns related to health, shaped by choices individuals make within the opportunities available to them based on their life circumstances. According to Health Lifestyle Theory, these choices are not isolated or random but tend to cluster in recognizable patterns influenced by factors like class, gender, and other structural variables. Formulated by Cockerham in 2005 and based on the earlier work of Weber and Bourdieu, the theory underscores how structural elements such as class, age, gender, race/ethnicity, social networks, and living conditions establish the social context that shapes socialization and experiences, ultimately influencing lifestyle tendencies and practices (Cockerham, 2020).

The Health Lifestyle Theory is used in this study to understand how lifestyle practices influence the respondents' vulnerability to respiratory diseases. This theory is a useful framework to know the reason behind the lifestyle of the respondents, which is the motivation of this study e.g., the influence of lifestyle to respiratory diseases. In this context, the health lifestyle theory helped in understanding how the specific lifestyle choices such as diet, physical activity, and hygiene within the correctional setting contribute to the vulnerability to respiratory diseases of the respondents. It delves into how behaviors such as hygiene, physical activity, or diet may collectively influence respiratory health. By applying the theory, the researchers can focus on the lifestyle aspects that directly contribute to vulnerability, providing a targeted perspective on the role of daily behaviors in shaping health outcomes among individuals in the correctional facility.

Another theory that was used in this study is the Theory of Planned Behavior of Ajzen. The theory of planned behavior states that when an individual plans to perform a given behavior, they will do so. The idea of planned behavior states that behavioral

intentions are established based on attitudes, subjective norms, and perceived behavioral control (Gu, 2019). These theories play a huge role in understanding the influence of lifestyle practices and the vulnerability to respiratory diseases of the respondents. Through these theories, the researchers were able to understand how the health lifestyles of the respondents interplay, contributed by their attitudes, subjective norms, and perceived behavioral control, ultimately leading to their level of vulnerability to respiratory diseases.

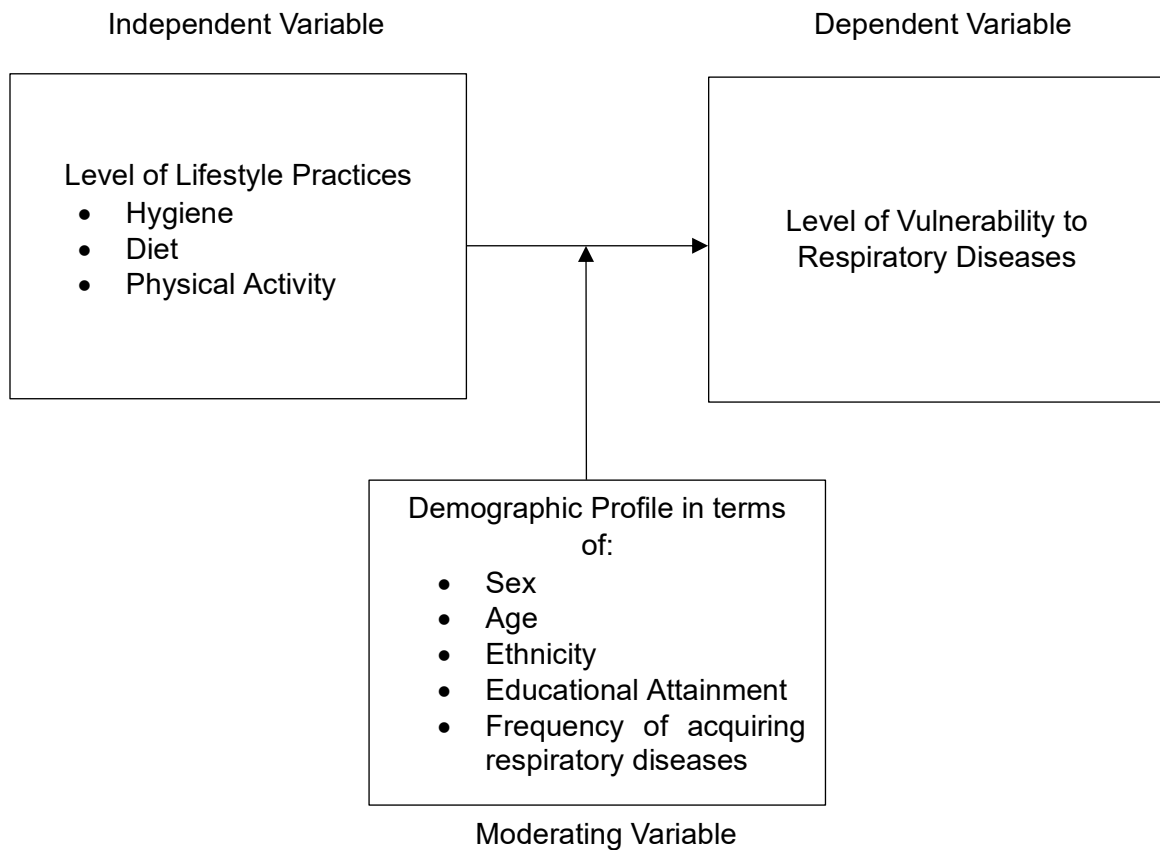
Conceptual Framework

The Cockerham health lifestyle theory is the foundation for the study's conceptual framework. The independent and dependent variables in the study will correlate, as explained by the hypothesis. The independent variable is the influence of lifestyle practices (diet, hygiene, physical activity) and the vulnerability to respiratory diseases of the respondents. On the other hand, the dependent variable pertains to the vulnerability to respiratory diseases of the respondents.

The conceptual framework shows a direct arrow from the independent variables to the level of vulnerability to respiratory diseases. It means the three variables impact or affects the respondents' vulnerability to respiratory diseases. Given that lifestyle might influence the respondents' vulnerability to respiratory diseases when they are in prison.

Moderating factors, including sex, ethnicity, and educational attainment, are all included in the framework. According to these moderating variables, the correlation between respondents' lifestyle practices and vulnerability to respiratory diseases may be moderated by their demographic profile. It provides a thorough insight of how their lifestyle behaviors may vary depending on their demographic profile.

Figure 1. Conceptual Framework Showing the Variables of the Study



Definitions of Terms

The following terms used in this study are operationally defined as follows:

Lifestyle - refers to the way of life for an individual or group as determined by their social and economic circumstances, personal choices, and cultural norms. This includes aspects such as diet, exercise, sleep, substance use, and social interactions.

Vulnerability - refers to the degree to which an individual or group is susceptible to harm, including physical, mental, and emotional harm. Vulnerability can be influenced by various factors, including age, health status, socioeconomic status, and social support networks.

Persons Deprived of Liberty (PDL) - refers to the individuals who are restricted from their freedom of movement, such as prisoners, detainees, and those held in immigration detention centers.

Respiratory Diseases - refers to any of the diseases and disorders of the airways and the lungs that affect human respiration.

Davao Oriental Provincial Jail - refers to a correctional facility located in Davao Oriental, Philippines, that houses PDLs awaiting trial or serving sentences.

CHAPTER III

METHODOLOGY

In this chapter, the methodology is used to carry out the study. It consists of different methods that was used by the researchers. Included are descriptions of the research design, research locale, research participants, research instruments, data collection procedure, and ethical considerations.

Research Design

This study was considered quantitative research because it qualified the specifications of quantitative research: collecting and analyzing numerical data (Bhandari, 2020). Specifically, the descriptive correlational and comparative research designs were utilized, as they were most appropriate for the nature and objectives of this study.

Furthermore, the researchers utilized the descriptive correlational research design, which was a research approach aimed at understanding the connection between two or more variables without asserting causation. It involved gathering and examining data on at least two variables to identify potential relationships (Bhat, 2020). This design was appropriate for the study's objective of identifying whether there was a significant relationship between the respondents' lifestyle and their level of vulnerability to respiratory diseases.

This study also used a comparative research design. The comparative approach was commonly utilized to identify and analyze patterns of similarities and differences, providing insights into the explanations for both continuity and change (Nupi, n.d.). This

design directly supported the research question which aimed to determine if there was a significant difference in the respondents' vulnerability to respiratory diseases when grouped according to their demographic profile.

Therefore, the use of both descriptive correlational and comparative designs was well-suited for the study's purpose of helping the researchers describe the characteristics of the study population, compare the vulnerability of respondents when grouped according to demographic profile, and examine the relationships between respondents' lifestyle and vulnerability to respiratory diseases.

Research Locale

The study site of this study was in Davao Oriental Provincial Jail. It is located in Sitio Tagbobolo, Barangay Sainz, Mati City, Davao Oriental. It is headed by the Provincial Jail Warden, Mr. Apollo Conde. The correctional facility is 1.8 km from the national highway with a total land area of 5,500 sq. meter. The facility has a total population of four hundred thirty-four (434) persons deprived of liberty and has nine (9) cells that can each accommodate sixty-two (62) PDLs. The researchers decided to conduct the study in this area as it is the only large-scale jail in the province.

Figure 2. Spot Map of Davao Oriental Provincial Jail
Retrieved from Google Maps



Research Participants

The respondents of this study are the persons deprived of liberty in Davao Oriental Provincial Jail. This study had sixty (62) respondents. The inclusion criteria specify that the respondents must be aged 18 to 65, currently incarcerated, has the willingness to participate in the study and provide informed consent, and is part of the cluster 3. On the other hand, the exclusion criteria are comprised of individuals below the age of 18 and above 65, those who have never been incarcerated, lack of willingness or ability to provide informed consent, individuals with a known history of severe mental illness or cognitive impairment that may affect their ability to participate in the study, and those individuals already diagnosed with respiratory diseases.

The Davao Oriental Provincial Jail has a total population of four hundred thirty-four (434) persons deprived of liberty. Due to the restricted nature of the research setting and ethical considerations surrounding access to persons deprived of liberty (PDLs), the study employed convenience sampling. The sample size was determined based on the maximum number of participants permitted by the jail administration. Upon securing approval from the facility's warden, we were authorized to survey 62 respondents, who were selected based on their availability and willingness to participate during the data collection period.

While convenience sampling may limit the generalizability of findings, this approach was necessary given the institutional constraints. Nevertheless, efforts were made to ensure a diverse representation of the PDLs population within the permitted sample.

Research Instrument

The study's data was gathered using an adapted survey questionnaire. The questionnaire helped the researchers determine the level of lifestyle practices and the level of vulnerability to respiratory diseases of the respondents. The questionnaire was created by combining several validated tools, including KomPAN (2020), which was

used in a study by Al-Rifaai et al. (2018), the internationally recognized International Physical Activity Questionnaire (IPAQ, 2002), and a self-assessment tool used by the University of Cambridge in 2022. There were 18 items with three constructs, namely (1) Hygiene, (2) Diet and (3) Physical Activity.

The Hygiene construct examines how the respondents maintain personal and environmental hygiene, identifying and evaluating variables that either increase or decrease the risk of respiratory infections. In order to find relationships between dietary choices and respiratory health outcomes, the Diet construct examines the eating patterns of the respondents. Finally, the Physical Activity construct examines the kinds and quantities of physical activities that the respondents engage in, providing insight into the possible effects of exercise on respiratory health.

This study's adapted questionnaire is divided into three sections, each of which played a specific and vital role in revealing the level of lifestyle practices and the level of vulnerability to respiratory diseases of the respondents. The first section of the questionnaire focused on gathering the demographic profile of the respondents. It gathered important information such as age, sex, and educational attainment. The second section is intended to study the independent variable, which is the respondents' lifestyle within the prison. This section of the questionnaire sought to thoroughly investigate multiple aspects of the respondents' daily routines, habits, and practices, which collectively comprise their lifestyle. This section may contain topics like hygiene practices, dietary habits, and the kind and extent of physical activities in prison. The questionnaire's third and last section focused on the dependent variable, which addressed the respondent's vulnerability to respiratory diseases. This section may give helpful information on the possible relationships between lifestyle variables and respiratory health outcomes in the incarcerated population.

The respondents were requested to give their answers for each item in the first section using a 5-point Likert-type scale of 1 (poor), 2 (fair), 3 (good), 4 (very good) and 5 (excellent). In this questionnaire, the highest score in each factor indicated that the behavior mentioned has been exhibited at all times.

In the second section of the questionnaire, the respondents were requested to give their answer using a 5-point Likert-type scale of 1 (strongly agree), 2 (agree), 3 (neutral), 4 (disagree) and 5 (strongly disagree). In this questionnaire, the lowest score in each factor indicated that the respondent is extremely high vulnerable to respiratory diseases.

The researchers adapted a questionnaire that suited the study and was submitted to the panel of expert for validation. The Cronbach's Alpha was utilized to perform the reliability test during the questionnaire's pilot testing. The results are shown in the table below:

Table 1. *Cronbach's Alpha Results*

Variables	Cronbach's Alpha	N of Items	Level
Hygiene	.781	6	Acceptable
Diet	.804	6	Good
Physical activity	.792	6	Acceptable
Vulnerability	.749	6	Acceptable
Overall questionnaire	.833	24	Good

The overall questionnaire yielded a Cronbach's Alpha of 0.833, indicating good reliability. Among the variables, diet had the highest reliability score (0.804), followed by physical activity (0.792) and hygiene (0.781), all within acceptable to good ranges. This indicate that all variables and the overall questionnaire have an alpha value exceeding 0.7, signifying internal consistency and that the questionnaire passes the reliability test. With this, the researchers were able to use the questionnaire for their main respondents.

Moreover, the data collection was conducted on the second semester of the academic year 2024- 2025.

The following scale was used to evaluate the level of lifestyle practices of the respondents.

Table 2. *Parameter to evaluate the level of lifestyle practices of the respondents*

Score	Scale	Response	Description	Interpretation
5	4.21 - 5.0	Always	Excellent	The behavior mentioned has been exhibited at all times.
4	3.41 - 4.20	Often	Very Good	The behavior mentioned has been exhibited often
3	2.61 - 3.40	Sometimes	Good	The behavior mentioned has been sometimes exhibited.
2	1.81 - 2.60	Rarely	Fair	The behavior mentioned has been exhibited rarely.
1	1.0 – 1.80	Never	Poor	The behavior mentioned has not been exhibited.

In evaluating the level of vulnerability to respiratory diseases of the respondents, the following scale was used.

Table 3. *Parameter to evaluate the level of vulnerability to respiratory diseases of the respondents*

Score	Scale	Response	Description	Interpretation
5	4.21 - 5.0	Strongly Agree	Extremely Low Vulnerability	The PDL may have zero to low vulnerability to respiratory disease.
4	3.41 - 4.20	Agree	Low Vulnerability	The PDL may have low vulnerability to respiratory disease.
3	2.61 - 3.40	Neutral	Moderate Vulnerability	The PDL may have moderate vulnerability to respiratory disease.
2	1.81 - 2.60	Disagree	High Vulnerability	The PDL may have high vulnerability to respiratory disease.
1	1.0 – 1.80	Strongly Disagree	Extremely High Vulnerability	The PDL may have extreme vulnerability to respiratory disease.

Data Collection Procedure

Data collection is the process of collecting data aiming to gain insights regarding the research topic (Taherdoost, 2021). The researchers collected the data in the following ways:

1. The researchers submitted a formal request letter directed to the Provincial Jail Warden of Davao Oriental, seeking permission to conduct research, assistance, and security. This letter included specific information such as the total population of persons deprived of liberty for the period 2024–2025, the total number of cells

within the facility, the capacity of each cell, total land area, and a comprehensive map of the entire jail premises. Once the request was approved, the researchers obtained the formal approval acknowledgment.

2. The researchers sought permission to access information regarding the total population of persons deprived of liberty (PDLs) within the designated research area to determine the sampling method to utilize.
3. The researchers used an adapted questionnaire, subjected to a reliability test through pilot testing and Cronbach Analysis (0.833) overseen by the University Research Ethics Board. The questionnaire was formulated using pertinent questions adapted from relevant studies. Once approved by the thesis adviser, it was ready for distribution among individuals incarcerated at Davao Oriental Provincial Jail.
4. The researchers utilized an adapted Likert Scale survey questionnaire with corresponding descriptions and interpretations for each response option.
5. The researchers adhered to the agreed-upon schedule arrangement provided for conducting the data-gathering process.
6. Respondents were selected through convenience sampling. The research involved 62 persons deprived of liberty as respondents. For security reasons, the data collection spanned half a day by batch, consisting of 3 respondents answering the questionnaire in the assigned area with security. The time of collecting the data was from 9:00 AM to 12:00 PM, as it was the free time of the respondents.
7. The next step involved distributing the prepared materials to the chosen respondents at Davao Oriental Provincial Jail. The respondents utilized crayons to complete the questionnaire.
8. Respondents were escorted to the assigned area where they comfortably

answered the questions while ensuring their privacy and confidentiality. This place also had security personnels for the safety of the researchers and only 3 respondents were allowed to answer the questionnaire at the same time. The researchers provided a detailed explanation of the study before data collection commenced. If there were questions the respondents preferred not to answer, they skipped them and proceeded to the next. Additionally, the researchers verbally translated the questions into the local language to facilitate comprehension for those who may not understand Standard English or those who cannot read.

9. The researchers provided the questionnaires to the respondents. They then explained the purpose of the study and guided the respondents through the consent process. If they agreed, they proceeded to answer the questions. Upon completing the survey questionnaire, respondents were required to deposit it into the provided drop box. This process was designed to safeguard the privacy and confidentiality of their answers.
10. After that, the researchers provided a token of appreciation to show their gratitude to the respondents.
11. The data were collected and tabulated, requiring the expertise of a statistician. Once the data were tabulated, the researchers placed the answered survey questionnaires into a secure box inaccessible to others. The researchers then obtained a letter that the statistician signed.
12. The final manuscript was written and was presented to the technical panel where the result of the research was discussed.
13. The research findings will be published through a respected peer-reviewed journal or platform. Following the completion of the study, the gathered data will be securely disposed of, either through burning in the case of hard copies or

deletion for digital files, exemplifying a dedication to ethical research principles and protecting respondent anonymity.

14. Respondents who desire to view and obtain a copy of the results will be accommodated. They can reach out to the researchers via email or cell phone.

15. Since this information can serve as baseline data for developing a program to mitigate the vulnerability to respiratory diseases of the respondents, the Davao Oriental Provincial Jail and the Faculty of Nursing and Allied Health Sciences will receive a complete copy of the study.

Ethical Consideration

The study was carried out with adherence to research principles and ethics. In addition to the letter attached to the questionnaire, respondents received a thorough explanation of the study before data collection begins.

Inclusion and Exclusion Criteria. The inclusion criteria specify that the respondents must be aged 18 to 65, currently incarcerated, have the willingness to participate in the study and provide informed consent, and be part of cluster 3. On the other hand, the exclusion criteria include individuals below the age of 18 and above 65, those who have never been incarcerated or those in the facility for less than a month, lack of willingness or ability to provide informed consent, with a known history of severe mental illness or cognitive impairment that may affect their ability to participate in the study, and those already diagnosed with respiratory diseases.

Involvement of Vulnerable Respondents. This study involved respondents who are considered a vulnerable population due to their restricted autonomy and potential exposure to coercion (Prisoners in Research Studies | Human Research Protection Program (HRPP), n.d.). Extra precautions were taken to ensure that participation is entirely voluntary and that informed consent is obtained without undue influence.

Researchers worked closely with jail authorities to guarantee ethical compliance while safeguarding participants' rights. Confidentiality and anonymity were strictly upheld to minimize risks associated with participation. Furthermore, additional measures were implemented to ensure that the respondents understand the study's objectives, risks, and benefits, thereby promoting their autonomy and ethical inclusion in the research.

Minimization of Risk. Before administering the questionnaire, the respondents were asked to sign an Informed Consent Form. This process ensures that the respondents are fully informed about the study's purpose, procedures, and their rights, aligning with ethical guidelines for human research (NHMRC, 2018). The respondents were assured of confidentiality and informed that participation is entirely voluntary, allowing them to withdraw at any time without explanation. To prevent a breach of confidentiality, respondents' names were not written on the questionnaire, adhering to best practices outlined by the NHMRC (2018) regarding the protection of respondents' privacy in research settings.

Privacy and Confidentiality. In alignment with the American Psychological Association's guidelines, the respondents were assured that their responses will be kept confidential, following Principle B: Fidelity and Responsibility. This means that only the research team will have access to the data, in accordance with Standard 4.01:

Maintaining Confidentiality. By prioritizing the respondents' privacy, the researchers uphold ethical standards and build trust in the research process. In addition, the respondents were assured that their responses will remain confidential, in accordance with the health information privacy standards outlined by the Department of Health and Human Services (2014). These standards, often associated with the Health Insurance Portability and Accountability Act (HIPAA), emphasize the importance of safeguarding respondents' personal health information. By following these guidelines, the research team ensures that the respondent data is handled securely and only accessible to

authorized personnel. This commitment to privacy protection aligns with the ethical principles of confidentiality in research, fostering trust and compliance among the respondents.

Voluntary Participation. Participation in this study is entirely voluntary. The respondents received a clear and comprehensive information about the study's purpose, procedures, risks, and benefits before providing their consent. They have the right to withdraw at any stage without facing any consequences. Informed consent was obtained either in writing or verbally, depending on the literacy level of the respondents. Special attention will be given to ensure that respondents do not feel pressured or coerced into participating.

Minimizing Harm to Living Beings and Environmental Conservation. In this study, the researchers are dedicated to upholding ethical guidelines that prioritize the well-being of living beings and environmental conservation. To achieve this, the researchers ensured that research activities are conducted in a manner that respects the dignity and rights of all living beings involved. This includes taking measures to minimize noise pollution, refrain from causing harm to wildlife or vegetation, and implement waste management practices to reduce environmental impact. By adhering to these ethical guidelines, the researchers conducted the research in a manner that promotes harmony with the environment and demonstrates our commitment to responsible and sustainable practices (United Nations, 2015).

Validity and Reliability. The research questionnaire was subjected to pilot testing and Cronbach Analysis to ensure the validity and reliability of the research instrument. Pilot testing involves administering the questionnaire to a small sample of respondents to identify any potential issues with clarity, comprehension, or relevance of the questions. Additionally, Cronbach's analysis was conducted to assess the internal consistency and reliability of the questionnaire items. The results of these tests are

crucial for proving the reliability and validity of the research instrument, which enhances the credibility of the study's findings (Bryman, 2016).

CHAPTER IV

RESULTS AND DISCUSSIONS

The obtained data from the respondents regarding their demographic profile, respiratory health, and lifestyle practices are presented, analyzed, and interpreted in this section based on the research objectives previously stated. The order of the discussion is as follows: the respondents' demographic profile, frequency of respiratory illnesses, and lifestyle practices in terms of hygiene, diet, and physical activity, as they relate to vulnerability to respiratory diseases among the respondents.

Data Analysis

The survey data have undergone analysis using descriptive statistics, such as frequencies, percentages, and means, to provide an overview of the demographic characteristics of the respondents and their level of vulnerability. These statistical measures will establish a baseline for the sample's characteristics.

In this study, the statistical tools that were used in analyzing and interpreting the gathered data were the following:

- Frequency and Percentage were used to categorize respondents based on their demographic profile (Sex, Age, Ethnicity, Educational Attainment, Frequency of acquiring respiratory diseases) in the first part of the questionnaire.
- Below is the used formula:

$$P = \frac{F}{N} \times 100$$

Where:

P = percentage

F = frequency

N = Number of respondents

- Mean was used to determine the level of lifestyle practices (hygiene, diet, and physical activity) of the respondents and the level of vulnerability to respiratory diseases of the respondents.
- Pearson correlation coefficient was used to examine and measure the relationship between the level of lifestyle practices (including hygiene, diet, and physical activity) and the level of vulnerability to respiratory diseases among the respondents.
- Analysis of Variance (ANOVA) is used to compare the respondents' vulnerability to respiratory diseases when grouped according to demographic profile.
- Multiple regression analysis was used to determine if lifestyle practices in terms of hygiene, diet, and physical activity predict the level of vulnerability to respiratory diseases of the respondents.

Table 4. *Sociodemographic Profile of the Respondents*

Demographic Profiles		Frequency (n = 62)	Percent (100%)
Age	18-25	7	11.3
	26-35	21	33.9
	36-45	21	33.9
	46-55	11	17.7
	56-65	2	3.2
Sex	Female	12	19.4
	Male	50	80.6
Educational Attainment	College Graduate	4	6.5
	College Level	8	12.9
	High School Graduate	15	24.2
	High School Level	17	27.4
	Elementary Graduate	6	9.7
	Elementary Level	12	19.4
Ethnicity	Bisaya	36	58.1
	Bul-anon	1	1.6
	Kalagan	11	17.7
	Mandaya	11	17.7
	Mansaka	2	3.2
	Mindanaon	1	1.6
Experienced Respiratory Illness in the Past Year	1 to 3 times	31	50.0
	10 or more	4	6.5
	4 to 6 times	8	12.9
	7 to 9 times	2	3.2
	Never	17	27.4

Table 4 presents the analysis of the demographic profile. It reveals that the majority of respondents were aged 26–45 (67.8%), aligning with the Bureau of Jail Management and Penology (2022), which states that the age distribution of most Persons Deprived of Liberty (PDLs) falls within the 18–59 age range. Specifically, the age group with the highest number of PDLs in Region XI is the 25–30 age bracket.

Furthermore, in terms of sex, males significantly outnumbered females, comprising 80.6% of respondents, while females accounted for only 19.4%. This result is

consistent with global data indicating that 93% of the incarcerated population are men (Data Development and Dissemination Section et al., 2021). Although men dominate prison populations, the rising rate of female incarceration is noteworthy.

In addition, high school education was the most common attainment level, with respondents either reaching the high school level (27.4%) or graduated (24.2%). This supports Guadamor (2020), who reported that high school was the highest educational level achieved by most inmates in their study.

Moreover, in terms of ethnicity, the majority were Bisaya (58.1%), followed by Kalagan (17.7%) and Mandaya (17.7%). This reflects the demographic composition of Region XI, where Bisaya is the dominant ethnic group (PSA, 2023). Davao Oriental, in particular, has been home to Indigenous peoples since ancient times, including the Mandaya, Mansaka, Mamanua, Manobo, Mangguangan, Tagacaulo, and Kalagan or Kagan (Provincial Government of Davao Oriental, n.d.). The data, therefore, reflect the strong presence of Bisaya and Mandaya populations, indicating that the study area is predominantly based in Mindanao.

With regard to health, 50% of respondents reported experiencing respiratory illnesses one to three times in the past year. This result parallels with the finding of Eze et al. (2022), in which they state that 69.4% of respondents in their study reported at least one respiratory symptom.

Table 5. *Level of lifestyle practices in terms of Hygiene*

Statement	Mean	Description
1. I change into clothes daily.	5.00	Excellent
2. I wash my clothes at least once a week.	4.95	Excellent
3. I wash my hands with water and soap before eating and after using the toilet.	4.89	Excellent
4. I brush my teeth at least once daily.	4.89	Excellent
5. I cut my nails at least once every 2 weeks.	4.81	Excellent
6. I replace my toothbrush every 3 months.	4.71	Excellent
Overall mean	4.89	Excellent

The data in Table 5 indicate that the respondents demonstrated excellent level of lifestyle practice in terms of hygiene, with an overall mean score of 4.89. The highest score (5.00) was recorded for daily changing of clothes, while other practices such as handwashing and tooth brushing also scored highly. Even the lowest-rated item, which was replacing toothbrushes every three months, received a strong score of 4.79.

These findings are consistent with Engdaw et al. (2023), who reported generally good hygiene among inmates, with 54.3 percent demonstrating adherence to proper hygiene practices. This aligns with the present study, which also indicates excellent hygiene practices among respondents, including high mean scores for daily clothing changes, handwashing, and tooth brushing. The consistency between both studies suggests that, despite the restrictive environment of a correctional facility, the respondents exhibit commendable hygiene habits.

Table 6. *Level of lifestyle practices in terms of Diet*

Statement	Mean	Description
1. In a week, I eat protein-rich foods such as fish, meat, and/or dairy.	4.97	Excellent
2. In a week, I eat vegetables and fruits such as carrots, mangoes, and/or leafy greens.	4.73	Excellent
3. In a week, I consume fried foods.	4.73	Excellent
4. In a week, I drink sweetened carbonated or carbonated beverages such as soda and juice.	4.60	Excellent
5. In a week, I drink at least 1 liter of water daily.	4.41	Excellent
6. In a week, I eat carbohydrate-rich foods such as rice, bread, and/or pasta.	3.56	Excellent
Overall mean	4.50	Excellent

Table 6 presents the respondents' level of lifestyle practice in terms of diet, which indicate an excellent level with an overall mean score of 4.50. The highest score (4.97) was recorded for the consumption of protein-rich foods such as fish, meat, and dairy, indicating adequate intake of essential nutrients. High scores were also noted for the consumption of fruits and vegetables (4.73), as well as fried foods (4.73) and sweetened beverages (4.60), suggesting a diet that includes both healthy and less healthy items. Although carbohydrate intake scored lower (3.56), it remained within the excellent range.

These results imply that the respondents maintain a relatively balanced diet, but moderation of fried and sugary foods is advised to support better respiratory and overall health. This is supported by Mommaerts et al. (2022), who emphasized that while prison meals are designed to be nutritionally adequate, access to commissary items can disrupt a balanced diet. In contrast, studies by Wondimu et al. (2023) and Fakhry (2020) reported inadequate prison diets in low-income countries, often lacking essential nutrients. Compared to those contexts, the findings in Table 5 reflect a relatively favorable nutritional status among the respondents.

Table 7. Level of lifestyle practices in terms of Physical Activity

Statement	Mean	Description
1. In a week, I walk for at least 10 minutes daily.	4.79	Excellent
2. In a week, I engage in moderate activities such as carrying light loads, gardening, and/or light exercises (not including walking).	4.41	Excellent
3. In a week, I spend at least 30 minutes daily doing moderate activities.	4.38	Excellent
4. In a week, I spend at least 5 hours sitting.	4.30	Excellent
5. In a week, I engage in vigorous activities such as heavy lifting, playing basketball, and/or Zumba.	3.54	Very Good
6. In a week, I spend at least 30 minutes daily doing vigorous activities.	3.49	Very Good
Overall mean	4.15	Excellent

Table 7 shows that the respondents' lifestyle practice in terms of physical activity is rated as excellent, with an overall mean score of 4.15. The highest mean score (4.79) was recorded for walking at least 10 minutes daily, indicating that this accessible form of physical activity is commonly practiced in the jail setting. Moderate activities such as gardening (4.41) and light exercise (4.38) also received excellent ratings. In contrast, more vigorous activities like heavy lifting (3.54) and playing basketball (3.49) were rated slightly lower, though still within the "very good" range.

These findings suggest that while respondents engage regularly in light to moderate physical activities, participation in more intense exercises is limited. This supports Baldonado et al. (2022), who identified basketball as a popular activity among inmates, and Mutz and Müller (2023), who reported that many prisoners engage in structured or informal exercise routines. However, the mean score of 4.30 for sitting five or more hours indicates a notable sedentary period. This is consistent with Liguori and Calella (2024), who noted that most incarcerated individuals fail to meet WHO physical activity recommendations, with over 75% leading predominantly sedentary lifestyles.

Table 8. *Descriptive summary on the level of lifestyle practices of the respondents*

	Mean	Description
Hygiene	4.89	Excellent
Diet	4.50	Excellent
Physical Activity	4.15	Excellent
Overall mean	4.51	Excellent

Table 8 presents a summary of the respondents' lifestyle practices in terms of hygiene, diet, and physical activity. The overall mean score of 4.51 indicates that the respondents maintain an excellent lifestyle.

Beginning with Table 3, the demographic profile showed that most respondents were age 26 to 45 years old, primarily male, with high school education, and belonging mostly to the Bisaya and Mandaya ethnic groups. Table 4 revealed that hygiene practices were the strongest aspect of their lifestyle, with a mean score of 4.89, highlighting consistent personal cleanliness. Table 5 showed that dietary practices were also excellent, with a mean of 4.50, suggesting a generally balanced intake of nutrients. Table 6 reported that physical activity, while still rated excellent with a mean of 4.15, received the lowest score among the three indicators.

Overall, the findings suggest that the respondents are able to maintain commendable hygiene, nutrition, and physical activity levels, which can positively support their respiratory and overall health. In a study conducted by Shao et al. (2021) shows that maintaining good physical activity levels and proper nutrition significantly supports overall health, including respiratory health. These lifestyle habits improve body function, physical and cognitive performance, and help prevent conditions like obesity, diabetes, and heart disease. The World Health Organization emphasizes that even moderate physical activity offers substantial health benefits and reduces mortality risk. Furthermore, combining regular exercise with a nutritious diet has been found to

strengthen the immune system and promote better health outcomes than focusing on either alone. These findings support the idea that consistent hygiene, nutrition, and physical activity habits can contribute positively to both respiratory and general well-being.

Table 9. Level of vulnerability to respiratory diseases of the respondents

Statement	Mean	Description
1. I have received all the recommended Philippine vaccinations and COVID vaccinations.	4.41	Extremely Low Vulnerability
2. I am involved in low-risk activities such as spending most of my time in a well-ventilated space with proper health and safety measures.	4.41	Extremely Low Vulnerability
3. I am involved in medium-risk activities such as having high number of different face-to-face contacts.	4.33	Extremely Low Vulnerability
4. I am at higher risk of becoming seriously unwell from respiratory infection.	3.54	Low Vulnerability
5. I am involved in high-risk activities where I could be exposed to infection such as direct contact with COVID or TB-positive individuals.	3.19	Moderate Vulnerability
6. I have a health condition for which I have been offered or am eligible for new treatments for respiratory illness.	2.98	Moderate Vulnerability
Overall mean	3.81	Low Vulnerability

Table 9 shows that the respondents' level of vulnerability to respiratory diseases has an overall mean of 3.81, indicating a low vulnerability to respiratory diseases. The highest scores were recorded in vaccination status (4.41) and involvement in low- to medium-risk activities (4.33), suggesting that most respondents are vaccinated and generally not exposed to high-risk environments. This supports the findings of Gonçalves et al. (2021), who reported that the majority of detained individuals were classified as average risk, with only a small percentage falling into elevated or high-risk categories.

However, lower scores in the presence of pre-existing health conditions (2.98) and engagement in high-risk activities (3.19) highlight areas of concern. The moderate

score for perceived risk of serious illness (3.54) indicates that some respondents are aware of their susceptibility to respiratory issues. These results imply that while preventive measures like vaccination are effective, ongoing medical monitoring and targeted interventions for individuals with comorbidities remain necessary. This is supported by Altobelli et al. (2023), who found that age and existing health conditions significantly influence disease severity, and by Toofan et al. (2021), who linked multiple chronic illnesses to higher mortality risk.

Table 10. Relationship between the level of lifestyle practice and level of vulnerability to respiratory diseases of the respondents

Correlations		Hygiene	Diet	Physical Activity	Level of Lifestyle
Vulnerability	Pearson Correlation	-.030	.306*	.283*	.305*
	Sig. (2-tailed)	.819	.015	.026	.016

Table 10 shows the relationship between the respondents' level of lifestyle and their level of vulnerability to respiratory diseases. The data reveals a statistically significant moderate positive correlation between lifestyle and vulnerability, with a Pearson correlation value of 0.305 and a p-value of 0.016, which is less than the significance level of 0.05. This suggests that as the quality of lifestyle improves, the vulnerability to respiratory diseases tends to decrease. Among the specific lifestyle practice components, diet ($r = 0.306$, $p = 0.015$) and physical activity ($r = 0.283$, $p = 0.026$) also exhibit moderate positive correlations with vulnerability. In contrast, hygiene shows a very weak negative correlation with vulnerability ($r = -0.030$, $p = 0.819$), indicating almost no relationship between hygiene practices and vulnerability to respiratory diseases.

Table 11. *Comparison of Level of Vulnerability to Respiratory Diseases*

Demographic Profiles		Mean	F	P value
Age	18-25	3.79	0.931	0.452
	26-35	3.66		
	36-45	4.01		
	46-55	3.73		
	56-65	4.08		
Sex	Female	3.51	-1.977	0.053
	Male	3.90		
Educational Attainment	College Graduate	3.13	1.902	0.109
	College Level	4.10		
	Elementary Graduate	4.14		
	Elementary Level	3.93		
	High School Graduate	3.76		
	High School Level	3.71		
Ethnicity	Bisaya	3.85	0.271	0.927
	Bul-anon	3.33		
	Kalagan	3.82		
	Mandaya	3.77		
	Mansaka	4.00		
	Mindanaon	3.33		
Experienced Respiratory Illness in the Past Year	1 to 3 times	3.92	0.619	0.651
	10 or more	3.83		
	4 to 6 times	3.69		
	7 to 9 times	4.08		
	Never	3.66		

Table 11 presents the comparison of level of vulnerability to respiratory diseases across various demographic profiles of the respondents. The analysis shows that none of the demographic variables including age ($p = 0.452$), sex ($p = 0.053$), educational attainment ($p = 0.109$), ethnicity ($p = 0.927$), and experience of respiratory illness in the

past year ($p = 0.651$) had statistically significant differences in vulnerability levels, as all p -values are greater than the 0.05 significance level. The findings of this study contradict those of Häder et al. and Liu, Zheng, et al. (2022, 2023), who reported that older adults are more susceptible to lung infections and tend to experience more severe disease outcomes compared to younger individuals. They also emphasized that age is closely associated with disease progression and poor prognosis in individuals with COVID-19 infection.

Moreover, Reddy and Oliver (2023) emphasized the complexity of sex differences in the susceptibility, severity, and progression of chronic respiratory diseases. For instance, men exhibit a higher prevalence and worse outcomes in idiopathic pulmonary fibrosis (IPF), whereas women with chronic obstructive pulmonary disease (COPD) display distinct physiological patterns, experience more frequent hospitalizations, and show slower declines in mortality despite having better lung function metrics (Sodhi et al., 2022). Similarly, Silveyra et al. (2021) highlighted that sex-based differences in respiratory disease burden vary across the lifespan. Their findings indicate that conditions such as asthma, COPD, and lung cancer are often more severe or prevalent in females, while diseases like IPF, emphysema, chronic cough, and COVID-19 tend to be more severe in males—underscoring the influence of sex on disease manifestation at different life stages.

In addition, education level appears to play a protective role in respiratory health. Lan et al. (2024) reported that higher educational attainment may reduce the risk of respiratory diseases, partly by lowering smoking rates and body mass index, although much of the effect remains unexplained. Supporting this, Lee et al. (2022) found that lower education is associated with increased emphysema in adults with COPD. This relationship was even stronger with advancing age among individuals without COPD, though no significant association was found with airway wall thickness. These findings

suggest that educational disparities may impact disease severity through mechanisms beyond smoking and occupational exposures.

Finally, racial and ethnic disparities also emerge in respiratory health outcomes. Bhavnani et al. (2023) found disparities in the incidence of upper respiratory infections (URIs) across racial and ethnic groups, though not in the severity of illness. Ekström and Mannino (2022) further argued that race and ethnicity often serve as proxies for various social and environmental factors that contribute to poor health outcomes. While acknowledging that genetic differences between populations may exist, they noted that self-identified race or ethnicity can, to some extent, reflect underlying biological traits.

Table 12. *Regression Analysis of Lifestyle Practices as a Predictor of the Level Vulnerability to Respiratory Diseases*

Model	B	Std. Error	Beta	t	Sig.	R square
(Constant)	1.788	.817		2.187	.033	0.094
Diet	.449	.180	.306	2.494	.015	
Dependent variable: Vulnerability						

Table 12 presents the regression analysis on the predictors of vulnerability to respiratory diseases among the respondents. The p-value for diet is 0.015, which is less than the 0.05 significance level, indicating that diet is a significant predictor of vulnerability. Other lifestyle variables, such as hygiene and physical activity, were excluded from the model, indicating that they did not significantly contribute to predicting vulnerability in this study. The R squared value is 0.094, meaning that diet accounts for 9.4 percent of the variation in vulnerability levels.

This is in relation with the study of Salehi et al. (2024) which found that dietary habits, nutritional deficiencies, and specific nutrients — including vitamins D, E, and A, along with iron, zinc, selenium, magnesium, omega-3 fatty acids, and fiber — play a

significant role in preventing respiratory illnesses, alleviating symptoms, and reducing mortality rates. Their findings also highlight that fruit and vegetable consumption is associated with a lower risk of asthma and improved control, while dairy consumption may exacerbate symptoms and increase risk. Nutrients such as antioxidants, fats, fiber, and vitamin D are also believed to influence immune pathways related to asthma (Alwarith et al., 2020). Furthermore, poor-quality diets lacking sufficient high-quality protein, minerals, and fiber-rich fruits and vegetables are associated with an increased risk of chronic respiratory diseases, regardless of the dietary index used to assess diet quality (Li et al., 2024).

Scope and Limitation

This study focused on examining the relationship between the level of lifestyle practices, specifically hygiene, diet, and physical activity, and the vulnerability to respiratory diseases among the respondents. It involved 62 respondents selected through convenience sampling. Data were collected using an adapted questionnaire during the second semester of A.Y. 2024–2025. This study is limited to PDLs aged 18 to 65 who are housed in Cluster 3 of the facility, thereby excluding individuals outside this age range or assigned to other clusters. Furthermore, the results of this study may not be generalizable to a broader population due to the limited sample size and the non-random sampling method employed.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter provides a comprehensive summary of the study's key findings, examines their implications in the context of the research objectives, and offers well-founded recommendations to inform future actions.

SUMMARY

This study aimed to assess the relationship between the level of lifestyle practices, specifically hygiene, diet, and physical activity, and the vulnerability to respiratory diseases among the respondents. The findings are as follows:

A. Demographic profile of the respondents

Among the 62 respondents, the majority were aged between 26 to 45, making up 67.8% of the total. Regarding educational background, most respondents either reached the high school level (27.4%) or finished high school (24.2%). Ethnically, the Bisaya group comprised the majority at 58.1%. In terms of respiratory health, 50% of respondents reported experiencing respiratory illnesses 1 to 3 times in the past year.

B. Level of Lifestyle Practices

The respondents' overall lifestyle was rated as excellent, with a combined mean score of 4.51 across hygiene, diet, and physical activity. Hygiene received the highest rating (4.89), reflecting consistent personal cleanliness, which is vital for preventing respiratory illnesses. Diet followed closely with a mean of 4.50, indicating a generally balanced intake, though it includes frequent consumption

of fried and sugary foods. Physical activity scored the lowest among the three at 4.15, but still within the excellent range.

C. Level of vulnerability to respiratory diseases

The respondents' vulnerability to respiratory diseases was rated as low vulnerability, with an overall mean score of 3.81. This indicates generally steady health among most persons deprived of liberty, though some remain at risk.

D. Relationship between the level of lifestyle practices and level of vulnerability to respiratory diseases

The result reveals a significant correlation between the respondents' lifestyle and their vulnerability to respiratory diseases. A Pearson correlation of 0.305 ($p = 0.016$) indicates that better lifestyle quality is associated with lower vulnerability. Specifically, diet ($r = -0.306$) and physical activity ($r = 0.283$) were significantly linked to reduce risk, while hygiene showed no significant correlation ($r = -0.030$, $p = 0.819$).

E. Comparing vulnerability levels across demographic profiles

The analysis shows that none of the demographic variables including age ($p = 0.452$), sex ($p = 0.053$), educational attainment ($p = 0.109$), ethnicity ($p = 0.927$), and experience of respiratory illness in the past year ($p = 0.651$) had statistically significant differences in vulnerability levels, as all p-values are greater than the 0.05 significance level.

F. Regression analysis on the predictors of vulnerability among respondents

The p-value for diet is 0.015, which is less than the 0.05 significance level, indicating that diet is a significant predictor of vulnerability. Other lifestyle variables, such as hygiene and physical activity, were excluded from the model, indicating that they did not significantly contribute to predicting vulnerability in this study. The R squared value is 0.094, meaning that diet accounts for 9.4 percent

of the variation in vulnerability levels.

Conclusion

The findings revealed a statistically significant relationship between the respondents' lifestyle practices and their vulnerability to respiratory diseases. Thus, the null hypothesis (Ho1), stating that there is no significant relationship between the respondents' level of lifestyle practices and their level of vulnerability to respiratory diseases, is rejected. This implies that lifestyle practices, particularly diet and physical activity, are meaningfully related to the respondents' vulnerability to respiratory diseases. Meanwhile, the study failed to reject the null hypothesis (Ho2), indicating that there is no significant difference between the respondents' level of vulnerability to respiratory diseases when grouped according to demographic profile. This suggests that the vulnerability to respiratory diseases does not significantly vary based on age, sex, educational attainment, ethnicity, or past respiratory illness. Regression analysis further showed that only diet significantly predicted vulnerability, leading to the rejection of the null hypothesis (Ho3), which stated that lifestyle practices does not predict the level of vulnerability to respiratory diseases of the respondents. This highlights the key role of diet in respiratory health. The study emphasizes the importance of proper nutrition and physical activity in correctional settings and calls for continuous health monitoring. Additionally, the results support the relevance of Cockerham's Health Lifestyle Theory and Ajzen's Theory of Planned Behavior, affirming both as effective frameworks for understanding the link between lifestyle practices and respiratory disease vulnerability.

Recommendations

This quantitative research investigated the influence of lifestyle factors on the vulnerability of respondents to respiratory diseases, particularly within the Davao Oriental Provincial Jail. To address these concerns, it is recommended that regular and comprehensive health assessments be conducted to identify existing and potential health issues. Improving environmental conditions in the facility is also crucial to reducing health risks and promoting overall well-being, especially in areas where lifestyle practices and health vulnerabilities scored lower.

The following are categorized recommendations focused on nursing education, nursing practice, and nursing research to provide a holistic approach to improving the health of the respondents:

Nursing Education. Nursing education should cover the health challenges faced by respondents in jail, including how respiratory diseases spread, prevention strategies, health promotion methods effective in jail settings, and ethical considerations in caring for incarcerated individuals. Equipping future nurses with this knowledge fosters empathy and prepares them to care for vulnerable populations.

Davao Oriental Provincial Jail. The study results suggest practical steps to improve health and well-being within the jail. Management should implement programs promoting healthy eating, regular exercise, stress management, and hygiene. Providing nutritious meals, health education, hygiene supplies, clean surroundings, and access to healthcare services can help prevent illness and support a safer environment for both respondents and staff.

Nursing Practice. This study's findings may guide nurses in evaluating the health of respondents through thorough lifestyle assessments. It may also inspire the development of targeted interventions for those with respiratory diseases, considering their living conditions.

Future Researchers. Future research should explore a broader spectrum of health determinants within correctional institutions. Areas for further investigation include environmental contributors (e.g., air quality, overcrowding), stress-related health outcomes, and the quality and nutritional value of meals provided in detention facilities. Expanding knowledge in this area can inform better policies and promote fair healthcare practices in correctional facilities.

These recommendations offer a coordinated plan to improve the health of respondents. Through collaboration in nursing education, healthcare services, facility management, and research, sustainable strategies can be developed to reduce respiratory and other health risks. Promoting access to health education and care also supports a just and equitable approach to public health in correctional settings.

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APPENDIX A

ADVISORY AGREEMENT

**DAVAO ORIENTAL
STATE UNIVERSITY**

"A University of excellence, innovation, and inclusion"



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ADVISORY AGREEMENT

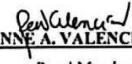
Faculty of Nursing and Allied Health Sciences (FNAHS)

We, the undersigned, hereby agree and formally express our commitment to constitute the Thesis Advisory Committee (TAC) of **FLORIELYN JOYCE P. APARECIO, ROSHABELLE C. ACUÑA, ELLAH MARIZ T. MAPA and KAYE P. VILLANIZO** in their pursuance to complete the requirements of the course, NCM 115 - Nursing Research 2, effective immediately upon approval by concerned authorities this 2st semester, SY 2023 to 2024.

Advisory Committee:


LENDELL KELLY B. YTAC, RN
Thesis Adviser

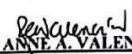

DR. REX S. APARICIO
Panel Member


RUTH ANNE A. VALENCIA, RN, MN
Panel Member


DR. MICHELLE SUZANNE L. TABOTABO, RN
Chair of the Panel

This duly signed composition of the *Thesis Advisory Committee (TAC)* is hereby endorsed and approved and shall be enforced until the completion of the study unless sooner revoked or terminated.

Endorsed by:


RUTH ANNE A. VALENCIA, RN, MN
Thesis Coordinator

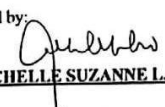
Date _____

Noted by:


DIANA VEM P. LAZARO, RN
Program Head

Date _____

Approved by:


DR. MICHELLE SUZANNE L. TABOTABO, RN
Dean

Date _____

APPENDIX B

PERMISSION TO CONDUCT STUDY

Republic of the Philippines

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LETTER OF PERMISSION TO CONDUCT STUDY

March 25, 2025

APOLLO CONDE
PROVINCIAL JAIL WARDEN
DAVAO ORIENTAL PROVINCIAL JAIL

Dear Sir:

Greetings of Peace and Joy!

We, the fourth-year Bachelor of Science in Nursing students, are conducting a study titled, "The Influence of Lifestyle on the Vulnerability of Persons Deprived of Liberty to Respiratory Diseases at Davao Oriental Provincial Jail".

In line with this, we respectfully seek your permission to conduct a survey among the persons deprived of liberty (PDLs) within Davao Oriental Provincial Jail. We are also asking if it is possible to conduct our study starting on March 27, 2025, or at a time most convenient for your institution.

We look forward to your favorable response to this request. Thank you very much.

Respectfully yours,

FLORIELYN JOYCE P. APARECIO

ROSHABELLE C. ACUÑA

ELCAH MARIZ T. MAPA

KAYE M. VILLANIZO

Approved by:

Lendell Kelly B. Ytac, RN
Thesis Adviser

Received by: **PJGA RYM FERNANDEZ**
3-25-2025

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APPENDIX C

STATISTICIAN CERTIFICATION FORM

Republic of the Philippines

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Certification of Statistical Analysis

This is to certify that the data presented in **"THE INFLUENCE OF LIFESTYLE ON THE VULNERABILITY OF PERSONS DEPRIVED OF LIBERTY TO RESPIRATORY DISEASES AT DAVAO ORIENTAL PROVINCIAL JAIL"** has undergone rigorous statistical analysis conducted by a certified statistician.

Certified Statistician: **ARJOHN M. CALIG-ONAN**

Date of Analysis: **April 5, 2025**

The statistical methods employed in this analysis are recognized and accepted within the statistical community for their accuracy and reliability.


ARJOHN M. CALIG-ONAN
Statistician

*Davao Oriental State University
Guang-guang, Dahican, City of Mati,
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APPENDIX D

GRAMMARIAN CERTIFICATION FORM

Republic of the Philippines

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GRAMMARIAN'S CERTIFICATE

This certifies that the undersigned has reviewed and gone through all pages of the thesis titled **"THE INFLUENCE OF LIFESTYLE PRACTICES AND THE VULNERABILITY TO RESPIRATORY DISEASES OF PERSONS DEPRIVED OF LIBERTY AT DAVAO ORIENTAL PROVINCIAL JAIL."** This thesis, authored by **FLORIELYN JOYCE P. APARECIO, ROSHABELLE C. ACUÑA, ELLAH MARIZ T. MAPA, and KAYE P. VILLANIZO**, aligns with the structural rules governing the composition of sentences, phrases, and words in the English language.

Signed this 19th day of June, 2025, at Mati City, Davao Oriental, Philippines.

Certified by:

A handwritten signature in black ink, appearing to read 'Haifa A. Basucan'.

Haifa A. Basucan, LPT
Grammarian

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APPENDIX E

SPOT MAP



APPENDIX F

INFORMED CONSENT

	DAVAO ORIENTAL STATE UNIVERSITY Research Ethics Office
DOrSU-REO Form 10 (A)	INFORMED CONSENT FORM
Version: 1	
Date of Effectivity: 02/09/2022	

Participant code: _____

Name of the Researcher(s): Florielyn Joyce P. Aparecio,
Roshabelle C. Acuña, Ellah Mariz T. Mapa, Kaye P.
Villanizo

Contact number: 09454163459

Title of the Study/project: The Influence of Lifestyle on the Vulnerability
of Persons Deprived of Liberty to Respiratory Diseases at
Davao Oriental Provincial Jail

INTRODUCTION

You are invited to participate in a study conducted by Florielyn Joyce P. Aparecio, Roshabelle C. Acuña, Ellah Mariz T. Mapa, Kaye P. Villanizo, at Davao Oriental Provincial Jail because you fit the inclusion criteria for informants of our study.

Your participation is completely voluntary. Please read the information below, and ask question about anything you do not understand, before decide to discuss participation with your family or friends.

If you decide to participate, you will be asked to sign this form and will be given a copy of this form.

OBJECTIVES OF THE STUDY

The objective of this study is to evaluate how lifestyle impacts the vulnerability of PDLs to respiratory diseases, aiming to address existing research gaps. Future studies should further investigate how lifestyle elements relate to vulnerability among those deprived of liberty. Recognizing the impact of lifestyle on vulnerability is pivotal in crafting precise interventions and support initiatives tailored to the unique requirements of PDL. Addressing the influence of lifestyle on their susceptibility is a crucial stride towards fostering rehabilitation, lowering reoffending rates, and ensuring equitable and efficient operations within the criminal justice system.

STUDY PROCEDURES

If you agree to participate, you will receive an orientation explaining the research's purpose, and we will obtain your informed consent. Subsequently, you will be asked to complete a survey questionnaire provided and collected by the researcher. You will be given a folder to shield your responses from other participants, and each participant will be coded to ensure identity anonymity. The survey questionnaire, will be deposited in a secure box to maintain the confidentiality of responses. The survey is designed to take only 10-15 minutes to complete.

If there are questions you prefer not to answer, you may skip them and proceed to the next. A designated area is available where you can comfortably respond to the survey. All information recorded is confidential, with your name excluded from the forms. Only the researcher will have access to your survey, ensuring the privacy of your responses.

POTENTIAL RISK AND DISCOMFORTS

You may feel discomfort during the course of the survey because of the sensitive nature of the topic being studied. You may opt not to answer questions which make you feel any psychological or emotional distress or you can withdraw as a participant of the study if you feel that you cannot discuss the information that is asked of you. The researchers value your participation and will place your welfare as their highest priority during the course of the study.

POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY

Findings from this study can inform the management of correctional facilities, enabling them to implement more effective measures to reduce disease transmission and improve healthcare within the jail. The research can contribute to evidence-based policymaking and the development of interventions tailored to the specific needs of PDLs. After the data gathering, the PDLs will be given a token of appreciation. This study is also instrumental in identifying the influence of lifestyle on the vulnerability of persons deprived of liberty to respiratory diseases.

CONFIDENTIALITY

We will keep your records for this study as far as permitted by law. Any identifiable information obtained in connection with this study will remain confidential, except if necessary to protect your rights or welfare. This certificate means that the researcher will not release of information about your information in publication or discuss in conferences and no identifiable information will be used.

PARTICIPATION AND WITHDRAWAL

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If you have any questions or concern about the research, please feel free to contact the researcher using the above information.

RIGHTS OF RESEARCH PARTICIPANT'S CONSENT

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+63 (087) 3883-195.

RESEARCH PARTICIPANT'S CONSENT

I have read the information provided above. I have been given a chance to ask questions and were answered to my satisfaction. I agree to participate in this study and is given a copy of this form. I understand that I can withdraw my consent at any time and discontinue participation without penalty.

Signature of Participant

Date Signed

To be accomplished by the Researcher Obtaining Consent:

I have explained the research to the participant and answered all of his/her questions. I believe that he/she understands the information described in this document and freely consents to participate.

Name of Person Obtaining Consent

Date Signed

APPENDIX G

SURVEY QUESTIONNAIRE

Part I. Demographic Profile.

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Sex: ☐ Female ☐ Male

Ethnicity: ☐ Bisaya ☐ Mandaya ☐ Kalagan ☐ Marana
o

☐ Mansaka ☐ Tagalog ☐ Ilonggo

Other, Pls. Specify: _____

Educational Attainment:

☐ Elementary Level ☐ Elementary Graduate

☐ High School Level

☐ High School Graduate

☐ College Level

☐ College Graduate

Age:

☐ 18-25

☐ 26-35

☐ 36-45

☐ 46-55

☐ 56-65

In the past year, how many times have you experienced respiratory illness such as cough, colds, or difficulty breathing?

☐ Never ☐ 4 to 6 times ☐ 10 or more

☐ 1 to 3 times ☐ 7 to 9 times

Part II. Assessing the lifestyle of persons deprived of liberty (PDLs).

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Always

The behavior mentioned has been exhibited at all times.

4 – Often

The behavior mentioned has been exhibited most of the time.

3 – Sometimes

The behavior mentioned has been exhibited sometimes.

2 – Rarely

The behavior mentioned has been exhibited rarely

1 – Never

The behavior mentioned has not been exhibited at all times.

A. HYGIENE	5	4	3	2	1
I wash my hands with water and soap before eating and after using the toilet.					
I cut my nails at least once every 2 weeks.					
I brush my teeth at least once daily.					
I replace my toothbrush every 3 months.					
I change into clothes daily.					
I wash my clothes at least once a week.					

B. DIET	5	4	3	2	1
In a week, I eat carbohydrate-rich foods such as rice, bread, and pasta at least five times.					
In a week, I eat protein-rich foods such as fish, meat, and dairy at least five times.					
In a week, I eat vegetables and fruits such as carrots, mangoes, and leafy greens at least five times.					
In a week, I consume fried foods more than three times.					

In a week, I drink sweetened carbonated or carbonated beverages such as soda and juice less than three times.					
In a week, I drink at least 1 liter of water daily.					

C. PHYSICAL ACTIVITY	5	4	3	2	1
In a week, I engage in vigorous activities such as heavy lifting, playing basketball, and zumba.					
In a week, I spend at least 30 minutes daily doing vigorous activities.					
In a week, I engage in moderate activities such as carrying light loads, gardening, and light exercises. Not including walking.					
In a week, I spend at least 30 minutes daily doing moderate activities.					
In a week, I walk for at least 10 minutes daily.					
In a week, I sit or remain inactive for more than five hours daily.					

Part III. The persons deprived of liberty's vulnerability to respiratory diseases.

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Stronly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

D. VULNERABILITY	5	4	3	2	1
I am at higher risk of becoming seriously unwell from respiratory infection					
I have a health condition for which I have been offered or eligible or new treatments for respiratory illness.					
I have NOT received all the recommended Philippine vaccinations and COVID vaccinations.					
I am involved in high risk activities where I could be exposed to infection such as in direct contact with known COVID or Tuberculosis positive individuals or with the virus.					
I am involved in medium risk activities such as having high number of different face-to-face contacts.					
I am involved in low risk activities such as spending most of my time in a well-ventilated space with proper health and safety measures in place.					

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

	DAVAO ORIENTAL STATE UNIVERSITY Research Ethics Office
DOrSU-REO Form 10 (A)	INFORMED CONSENT FORM
Version:1	
Date of Effectivity:02/09/2022	

Participant code: 40

Name of the Researcher(s): Florielyn Joyce P. Aparecio,
Roshabelle C. Acuña, Ellah Mariz T. Mapa,
Kaye P. Villanizo

Contact number: 09454163459

Title of the Study/project: The Influence of Lifestyle on the Vulnerability
of Persons Deprived of Liberty to Respiratory
Diseases at Davao Oriental Provincial Jail

INTRODUCTION

You are invited to participate in a study conducted by Florielyn Joyce P. Aparecio, Roshabelle C. Acuña, Ellah Mariz T. Mapa, Kaye P. Villanizo, at Davao Oriental Provincial Jail because you fit the inclusion criteria for informants of our study.

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OBJECTIVES OF THE STUDY

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ANSWERED SURVEY QUESTIONNAIRE

the impact of lifestyle on vulnerability is pivotal in crafting precise interventions and support initiatives tailored to the unique requirements of PDL. Addressing the influence of lifestyle on their susceptibility is a crucial stride towards fostering rehabilitation, lowering reoffending rates, and ensuring equitable and efficient operations within the criminal justice system.

STUDY PROCEDURES

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POTENTIAL RISK AND DISCOMFORTS

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POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY

Findings from this study can inform the management of correctional facilities, enabling them to implement more effective measures to reduce disease transmission and improve healthcare within the jail. The research can contribute to evidence-based policymaking and the development of interventions tailored to the specific needs of PDLs. After the data gathering, the PDLs will be given a token of appreciation. This study is also instrumental

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ANSWERED SURVEY QUESTIONNAIRE

in identifying the influence of lifestyle on the vulnerability of persons deprived of liberty to respiratory diseases.

CONFIDENTIALITY

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APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

SURVEY QUESTIONNAIRE

Part I. Demographic Profile.
 Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Sex: ☐ Female ☒ Male

Ethnicity: ☒ Bisaya ☐ Mandaya ☐ Kalagan ☐ Maranao
☐ Mansaka ☐ Tagalog ☐ Ilonggo
 Other, Pls. Specify: _____

Educational Attainment:

☐ Elementary Level ☐ Elementary Graduate
☐ High School Level
☒ High School Graduate
☐ College Level
☐ College Graduate

Age:

☐ 18-25
☐ 26-35
☒ 36-45
☐ 46-55
☐ 56-65

In the past year, how many times have you experienced respiratory illness such as cough, colds, or difficulty breathing?

☒ Never ☐ 4 to 6 times ☐ 10 or more
☐ 1 to 3 times ☐ 7 to 9 times

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

Part II. Assessing the lifestyle of persons deprived of liberty (PDLs).

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Always

4 – Often

3 – Sometimes

2 – Rarely

1 – Never

The behavior mentioned has been exhibited at all times.

The behavior mentioned has been exhibited most of the time.

The behavior mentioned has been exhibited sometimes.

The behavior mentioned has been exhibited rarely

The behavior mentioned has not been exhibited at all times.

A. HYGIENE	5	4	3	2	1
I wash my hands with water and soap before eating and after using the toilet.	✓				
I cut my nails at least once every 2 weeks.		✓			
I brush my teeth at least once daily.	✓				
I replace my toothbrush every 3 months.	✓				
I change into clothes daily.	✓				
I wash my clothes at least once a week.	✓				

B. DIET	5	4	3	2	1
In a week, I eat carbohydrate-rich foods such as rice, bread, and/or pasta.	✓				
In a week, I eat protein-rich foods such as fish, meat, and/or dairy.	✓				

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ANSWERED SURVEY QUESTIONNAIRE

In a week, I eat vegetables and fruits such as carrots, mangoes, and/or leafy greens.	5	4	3	2	1
In a week, I consume fried foods.	5	4	3	2	1
In a week, I drink sweetened carbonated or carbonated beverages such as soda and juice.	5	4	3	2	1
In a week, I drink at least 1 liter of water daily.	5	4	3	2	1

C. PHYSICAL ACTIVITY	5	4	3	2	1
In a week, I engage in vigorous activities such as heavy lifting, playing basketball, and/or zumba.	5	4	3	2	1
In a week, I spend at least 30 minutes daily doing vigorous activities.	5	4	3	2	1
In a week, I engage in moderate activities such as carrying light loads, gardening, and/or light exercises. Not including walking.	5	4	3	2	1
In a week, I spend at least 30 minutes daily doing moderate activities.	5	4	3	2	1
In a week, I walk for at least 10 minutes daily.	5	4	3	2	1
In a week, I spend at least an 5 hours sitting.	5	4	3	2	1

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ANSWERED SURVEY QUESTIONNAIRE

	DAVAO ORIENTAL STATE UNIVERSITY Research Ethics Office
DOrSU-REO Form 10 (A)	INFORMED CONSENT FORM
Version: 1	
Date of Effectivity: 02/09/2022	

Participant code: 56
 Name of the Researcher(s): Florielyn Joyce P. Aparecio,
Roshabelle C. Acuña, Ellah Mariz T. Mapa,
Kaye P. Villanizo
Contact number: 09454163459
 Title of the Study/project: The Influence of Lifestyle on the Vulnerability
of Persons Deprived of Liberty to Respiratory
Diseases at Davao Oriental Provincial Jail
INTRODUCTION

You are invited to participate in a study conducted by Florielyn Joyce P. Aparecio, Roshabelle C. Acuña, Ellah Mariz T. Mapa, Kaye P. Villanizo, at Davao Oriental Provincial Jail because you fit the inclusion criteria for informants of our study.

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If you decide to participate, you will be asked to sign this form and will be given a copy of this form.

OBJECTIVES OF THE STUDY

The objective of this study is to evaluate how lifestyle impacts the vulnerability of PDLs to respiratory diseases, aiming to address existing research gaps. Future studies should further investigate how lifestyle elements relate to vulnerability among those deprived of liberty. Recognizing

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

the impact of lifestyle on vulnerability is pivotal in crafting precise interventions and support initiatives tailored to the unique requirements of PDL. Addressing the influence of lifestyle on their susceptibility is a crucial stride towards fostering rehabilitation, lowering reoffending rates, and ensuring equitable and efficient operations within the criminal justice system.

STUDY PROCEDURES

If you agree to participate, you will receive an orientation explaining the research's purpose, and we will obtain your informed consent. Subsequently, you will be asked to complete a survey questionnaire provided and collected by the researcher. You will be given a folder to shield your responses from other participants, and each participant will be coded to ensure identity anonymity. The survey questionnaire, will be deposited in a secure box to maintain the confidentiality of responses. The survey is designed to take only 10-15 minutes to complete.

If there are questions you prefer not to answer, you may skip them and proceed to the next. A designated area is available where you can comfortably respond to the survey. All information recorded is confidential, with your name excluded from the forms. Only the researcher will have access to your survey, ensuring the privacy of your responses.

POTENTIAL RISK AND DISCOMFORTS

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POTENTIAL BENEFITS TO PARTICIPANTS AND/OR TO SOCIETY

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APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

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APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

SURVEY QUESTIONNAIRE

Part I. Demographic Profile.

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Sex: ☒ Female ☐ Male

Ethnicity: ☒ Bisaya ☐ Mandaya ☐ Kalagan ☐ Maranao
☐ Mansaka ☐ Tagalog ☐ Ilonggo

Other, Pls. Specify: _____

Educational Attainment:

☐ Elementary Level ☐ Elementary Graduate
☐ High School Level
☒ High School Graduate
☐ College Level
☐ College Graduate

Age:

☐ 18-25
☐ 26-35
☐ 36-45
☒ 46-55
☐ 56-65

In the past year, how many times have you experienced respiratory illness such as cough, colds, or difficulty breathing?

☐ Never ☐ 4 to 6 times ☐ 10 or more
☒ 1 to 3 times ☐ 7 to 9 times

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

Part II. Assessing the lifestyle of persons deprived of liberty (PDLs).

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Always

4 – Often

3 – Sometimes

2 – Rarely

1 – Never

The behavior mentioned has been exhibited at all times.

The behavior mentioned has been exhibited most of the time.

The behavior mentioned has been exhibited sometimes.

The behavior mentioned has been exhibited rarely

The behavior mentioned has not been exhibited at all times.

A. HYGIENE	5	4	3	2	1
I wash my hands with water and soap before eating and after using the toilet.	✓				
I cut my nails at least once every 2 weeks.	✓				
I brush my teeth at least once daily.	✓				
I replace my toothbrush every 3 months.	✓				
I change into clothes daily.	✓				
I wash my clothes at least once a week.	✓				

B. DIET	5	4	3	2	1
In a week, I eat carbohydrate-rich foods such as rice, bread, and/or pasta.	✓				
In a week, I eat protein-rich foods such as fish, meat, and/or dairy.	✓				

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

In a week, I eat vegetables and fruits such as carrots, mangoes, and/or leafy greens.		☒			
In a week, I consume fried foods.		☒			
In a week, I drink sweetened carbonated or carbonated beverages such as soda and juice.	☒				
In a week, I drink at least 1 liter of water daily.	☒				

C. PHYSICAL ACTIVITY	5	4	3	2	1
In a week, I engage in vigorous activities such as heavy lifting, playing basketball, and/or zumba.			☒		
In a week, I spend at least 30 minutes daily doing vigorous activities.			☒		
In a week, I engage in moderate activities such as carrying light loads, gardening, and/or light exercises. Not including walking.		☒			
In a week, I spend at least 30 minutes daily doing moderate activities.		☒			
In a week, I walk for at least 10 minutes daily.		☒			
In a week, I spend at least an 5 hours sitting.				☒	

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

Part III. The persons deprived of liberty's vulnerability to respiratory diseases.

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

D. VULNERABILITY	5	4	3	2	1
I am at higher risk of becoming seriously unwell from respiratory infection.		✓			
I have a health condition for which I have been offered or eligible or new treatments for respiratory illness.		✓			
I have received all the recommended Philippine vaccinations and COVID vaccinations.		✓			
I am involved in high-risk activities where I could be exposed to infection such as in direct contact with known COVID or Tuberculosis positive individuals or with the virus.		✓			
I am involved in medium risk activities such as having high number of different face-to-face contacts.		✓			
I am involved in low-risk activities such as spending most of my time in a well-ventilated space with proper health and safety measures in place.		✓			

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

	DAVAO ORIENTAL STATE UNIVERSITY Research Ethics Office
DOrSU-REO Form 10 (A)	INFORMED CONSENT FORM
Version:1	
Date of Effectivity:02/09/2022	

Participant code: 54
Name of the Researcher(s): Florielyn Joyce P. Aparecio,
Roshabelle C. Acuña, Ellah Mariz T. Mapa,
Kaye P. Villanizo
Contact number: 09454163459
Title of the Study/project: The Influence of Lifestyle on the Vulnerability
of Persons Deprived of Liberty to Respiratory
Diseases at Davao Oriental Provincial Jail
INTRODUCTION

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ANSWERED SURVEY QUESTIONNAIRE

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APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

[Signature] 3/27/25
Signature of Participant Date Signed

To be accomplished by the Researcher Obtaining Consent

I have explained the research to the participant and answered all of his/her questions. I believe that he/she understands the information described in this document and freely consents to participate.

Ellan Maria Mapa 3/27/25
Name of Person Obtaining Consent Date Signed

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

SURVEY QUESTIONNAIRE

Part I. Demographic Profile.
 Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Sex: ☐ Female ☒ Male

Ethnicity: ☒ Bisaya ☐ Mandaya ☐ Kalagan ☐ Maranao
☐ Mansaka ☒ Tagalog ☐ Ilonggo
 Other, Pls. Specify: _____

Educational Attainment:

☐ Elementary Level ☐ Elementary Graduate
☐ High School Level
☐ High School Graduate
☐ College Level
☒ College Graduate

Age:

☐ 18-25
☒ 26-35
☐ 36-45
☐ 46-55
☐ 56-65

In the past year, how many times have you experienced respiratory illness such as cough, colds, or difficulty breathing?

☐ Never ☐ 4 to 6 times ☐ 10 or more
☒ 1 to 3 times ☐ 7 to 9 times

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

Part II. Assessing the lifestyle of persons deprived of liberty (PDLs).

Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Always

4 – Often

3 – Sometimes

2 – Rarely

1 – Never

The behavior mentioned has been exhibited at all times.

The behavior mentioned has been exhibited most of the time.

The behavior mentioned has been exhibited sometimes.

The behavior mentioned has been exhibited rarely

The behavior mentioned has not been exhibited at all times.

A. HYGIENE	5	4	3	2	1
I wash my hands with water and soap before eating and after using the toilet.	✓				
I cut my nails at least once every 2 weeks.	✓				
I brush my teeth at least once daily.	✓				
I replace my toothbrush every 3 months.	✓				
I change into clothes daily.	✓				
I wash my clothes at least once a week.	✓				

B. DIET	5	4	3	2	1
In a week, I eat carbohydrate-rich foods such as rice, bread, and/or pasta.	✓				
In a week, I eat protein-rich foods such as fish, meat, and/or dairy.		✓			

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

In a week, I eat vegetables and fruits such as carrots, mangoes, and/or leafy greens.	☒	☐	☐	☐	☐
In a week, I consume fried foods.	☒	☐	☐	☐	☐
In a week, I drink sweetened carbonated or carbonated beverages such as soda and juice.	☒	☐	☐	☐	☐
In a week, I drink at least 1 liter of water daily.	☒	☐	☐	☐	☐

C. PHYSICAL ACTIVITY	5	4	3	2	1
In a week, I engage in vigorous activities such as heavy lifting, playing basketball, and/or zumba.	☐	☒	☐	☐	☐
In a week, I spend at least 30 minutes daily doing vigorous activities.	☐	☒	☐	☐	☐
In a week, I engage in moderate activities such as carrying light loads, gardening, and/or light exercises. Not including walking.	☐	☒	☐	☐	☐
In a week, I spend at least 30 minutes daily doing moderate activities.	☐	☒	☐	☐	☐
In a week, I walk for at least 10 minutes daily.	☒	☐	☐	☐	☐
In a week, I spend at least an 5 hours sitting.	☐	☐	☒	☐	☐

APPENDIX H

ANSWERED SURVEY QUESTIONNAIRE

Part III. The persons deprived of liberty's vulnerability to respiratory diseases.
 Instruction: Kindly put a check mark (✓) next to the information that applies to you.

Scale:

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

D. VULNERABILITY	5	4	3	2	1
I am at higher risk of becoming seriously unwell from respiratory infection.					✓
I have a health condition for which I have been offered or eligible or new treatments for respiratory illness.				✓	
I have received all the recommended Philippine vaccinations and COVID vaccinations.	✓				
I am involved in high-risk activities where I could be exposed to infection such as in direct contact with known COVID or Tuberculosis positive individuals or with the virus.					✓
I am involved in medium risk activities such as having high number of different face-to-face contacts.			✓		
I am involved in low-risk activities such as spending most of my time in a well-ventilated space with proper health and safety measures in place.			✓		

APPENDIX I

ETHICAL CLEARANCE

	DAVAO ORIENTAL STATE UNIVERSITY Research Ethics Office
DOrSU-UREB Form 31 Version: 1 Date of Effectivity: 02/09/2022	COMPLETED ETHICAL REVIEW

NAME OF PROPONENT/RESEARCHER: FLORIELYN JOYCE P. APARECIO, ROSHABELLE C. ACUNA, ELLAH MARIZ T. MAPA, KAYE P. VILLANIZO
Institution: DORSU-FNAHS
Address: GUANG-GUANG, BRGY. DAHICAN, CITY OF MATI, DAVAO ORIENTAL
Title of project/study: "The Influence of Lifestyle on the Vulnerability of Persons Deprived of Liberty to Respiratory Diseases at Davao Oriental Provincial Jail"
UREB Code: 2024-05-FJAparecio-Respiratory-Disease-FNAHS
Date of Manuscript Submission: May 05, 2025
UREB Comments/Need for Revisions: • No ethical issues found
Date of Completed Ethical Review: May 06, 2025
Ethical Notes for Publication/Final Defense: • Results of the study can now be presented to the thesis advisory committee for final defense. • Results of the study can be published.


EMILY S. ANTONIO
 UREB Director

APPENDIX J

Research Expenses

Materials	Cost
• Printing of 5 Manuscripts (250.00php)	1250.00php
• 5 pcs. Sliding Folders (10.00php)	50.00php
• 5 pcs. Paper Clips (10.00php)	50.00php
• 2 pcs. Crayons (25.00php)	50.00php
• 62 pcs Wings Detergent Liquid (5.50php)	341.00php
• 62 pcs. Colgate Toothpaste (5.00php)	310.00php
• 62 pcs. Gard Shampoo (5.50php)	341.00php
• 62 pcs. Bioderm Soap (6.00php)	372.00php
• 4 bags Biscuits (164.00php)	656.00php
• 2 bags Candy (34.00php)	68.00php
• 1 bag Paper Bag (134.00php)	134.00php
• Transportation (500.00php)	500.00php
• Statistician (1,500.00php)	1,500.00php
TOTAL:	5,622.00php

CURRICULUM VITAE



FLORIELYN JOYCE P. APARECIO

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Personal Data

Permanent Address:	Purok Bucana, Matiao, City of Mati, Davao Oriental
Date of Birth:	March 19, 2003
Place of Birth:	City of Mati, Davao Oriental
Age:	22
Sex:	Female
Height:	5'0
Weight:	45kg
Civil Status:	Single
Citizenship:	Filipino

Name of Parents

Father:	Froilan P. Aparecio
Mother:	Lucita P. Aparecio

Educational Attainment

Level	School Graduated	Year
Senior High School	Davao Oriental Regional Science High School	2021
Junior High School	Davao Oriental Regional Science High School	2019
Elementary	Gov. Leopoldo N. Lopez Sr. Memorial School	2015

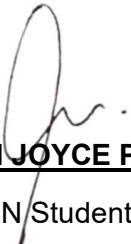
Affiliation

Organization	Position	Year
Red Cross Youth – Davao Oriental State University Council	Volunteer	2024

Trainings and Seminars

Trainings and Seminars Attended	Organization/Institute	Year
Standard First Aid Training	CDRRMO	2022
Emergency First Aid	Red Cross Youth – Davao Oriental State University Council	2024
Basic Life Support	Faculty of Nursing and Allied Health Sciences	2024

I certify that the above-mentioned data are all true and correct.


FLORIELYN JOYCE P. APARECIO

BSN Student

CURRICULUM VITAE



ELLAH MARIZ T. MAPA

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Personal Data

Permanent Address:	Purok Unity, Madang, Central, City of Mati, Davao Oriental
Date of Birth:	July 23, 2003
Place of Birth:	City of Mati, Davao Oriental
Age:	21
Sex:	Female
Height:	5'0
Weight:	40kg
Civil Status:	Single
Citizenship:	Filipino

Name of Parents

Father:	Jerry C. Mapa
Mother:	Generiza T. Mapa

Educational Attainment		
Level	School Graduated	Year
Senior High School	Davao Oriental Regional Science High School	2021
Junior High School	Davao Oriental Regional Science High School	2019
Elementary	Rabat Rocamora Mati Central SPED Center	2015

Affiliation		
Organization	Position	Year
Red Cross Youth – Davao Oriental State University Council	Volunteer	2024

Trainings and Seminars		
Trainings and Seminars Attended	Organization/Institute	Year
Emergency First Aid	Red Cross Youth – Davao Oriental State University Council	2024
Basic Life Support	Faculty of Nursing and Allied Health Sciences	2024

I certify that the above-mentioned data are all true and correct.


ELLAH MARIZ T. MAPA
 BSN Student

CURRICULUM VITAE



ROSHABELLE C. ACUÑA

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Personal Data

Permanent Address:	Purok Centro, Matiao, City of Mati, Davao Oriental
Date of Birth:	July 9, 2002
Place of Birth:	City of Mati, Davao Oriental
Age:	22
Sex:	Female
Height:	5'2
Weight:	54kg
Civil Status:	Single
Citizenship:	Filipino

Name of Parents

Father:	Nicodimos P. Acuña
Mother:	Rudith C. Acuña

Educational Attainment		
Level	School Graduated	Year
Senior High School	Davao Oriental Regional Science High School	2021
Junior High School	Davao Oriental Regional Science High School	2019
Elementary	Matiao Central Elementary School	2015

Affiliation		
Organization	Position	Year
Red Cross Youth – Davao Oriental State University Council	Volunteer	2024

Trainings and Seminars		
Trainings and Seminars Attended	Organization/Institute	Year
Emergency First Aid	Red Cross Youth – Davao Oriental State University Council	2024
Basic Life Support	Faculty of Nursing and Allied Health Sciences	2024

I certify that the above-mentioned data are all true and correct.


ROSHABELLE C. ACUÑA
 BSN Student

CURRICULUM VITAE



KAYE P. VILLANIZO

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Personal Data

Permanent Address:	San Jose Montevista, Davao de Oro
Date of Birth:	March 18, 2002
Place of Birth:	Montevista, Davao de Oro
Age:	23
Sex:	Female
Height:	5'5
Weight:	65kg
Civil Status:	Single
Citizenship:	Filipino

Name of Parents

Father:	Ireneo O. Villanizo
Mother:	Analyn V. Pedrezo

Educational Attainment

Level	School Graduated	Year
Senior High School	Montevista National High School	2021
Junior High School	Montevista National High School	2019
Elementary	Montevista Central Elementary School	2015

Affiliation

Organization	Position	Year
N/A		

Trainings and Seminars

Trainings and Seminars Attended	Organization/Institute	Year
Basic Life Support	Faculty of Nursing and Allied Health Sciences	2024

I certify that the above-mentioned data are all true and correct.


KAYE P. VILLANIZO
 BSN Student

GANTT CHART

	December 2023	January 2024	February to October 2024	November to February 2024	March 2025	April 2025	May 2025	June 2025
Research Title Consultation								
Revision of Concept Paper								
Thesis Proposal, Revision, and Routing								
Research Ethics Office								
Data Collection								
Data Analysis and Interpretation								
Presentation of Thesis								
Revision and Rerouting								
Final Defense								
Submission of Hard Bound Thesis								